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(54) **LIGHT EMITTING DEVICE, DISPLAY APPARATUS INCLUDING THE SAME, AND METHOD OF MANUFACTURING LIGHT EMITTING DEVICE**

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(57) **ABSTRACT**

The light emitting device includes a light emitting portion and an electrode portion each including an epi structure and separated from each other by a trench. The epi structure includes a first conductive semiconductor layer, an active layer, and a second conductive semiconductor layer which are sequentially stacked. A first contact layer and a second contact layer are respectively on the second conductive semiconductor layers of the light emitting portion and the electrode portion. A passivation layer is provided on side surfaces of the light emitting portion and the electrode portion, and the first contact layer and the second contact layer. An opening exposing the first contact layer of the light emitting portion is formed in the passivation layer. A conductive reflective layer is provided on the passivation layer and contacts the first contact layer of the light emitting portion through the opening. A plurality of pores are provided in the first conductive semiconductor layer.

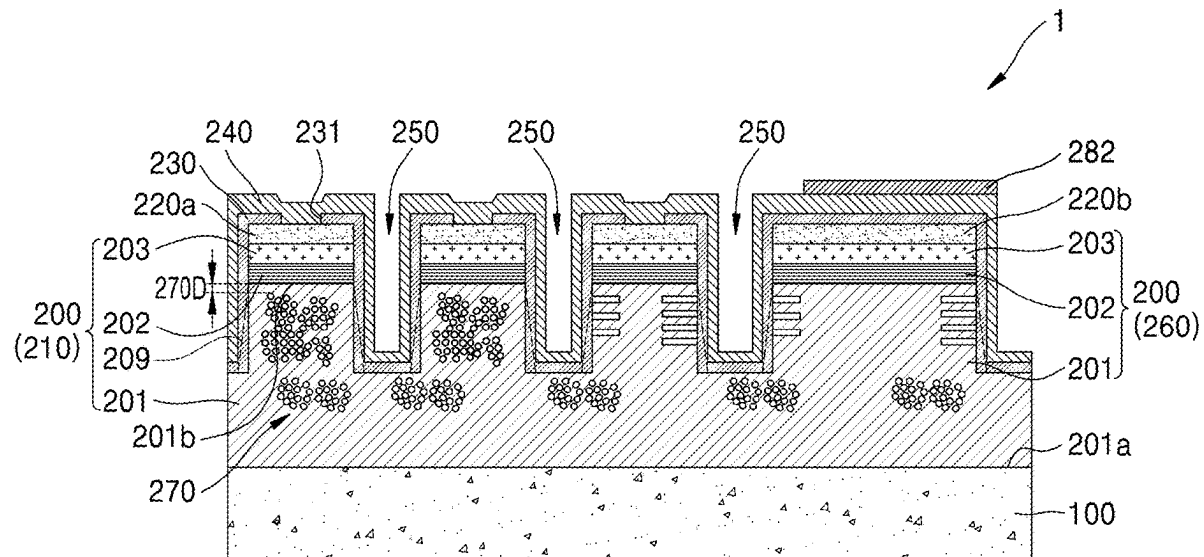


FIG. 2

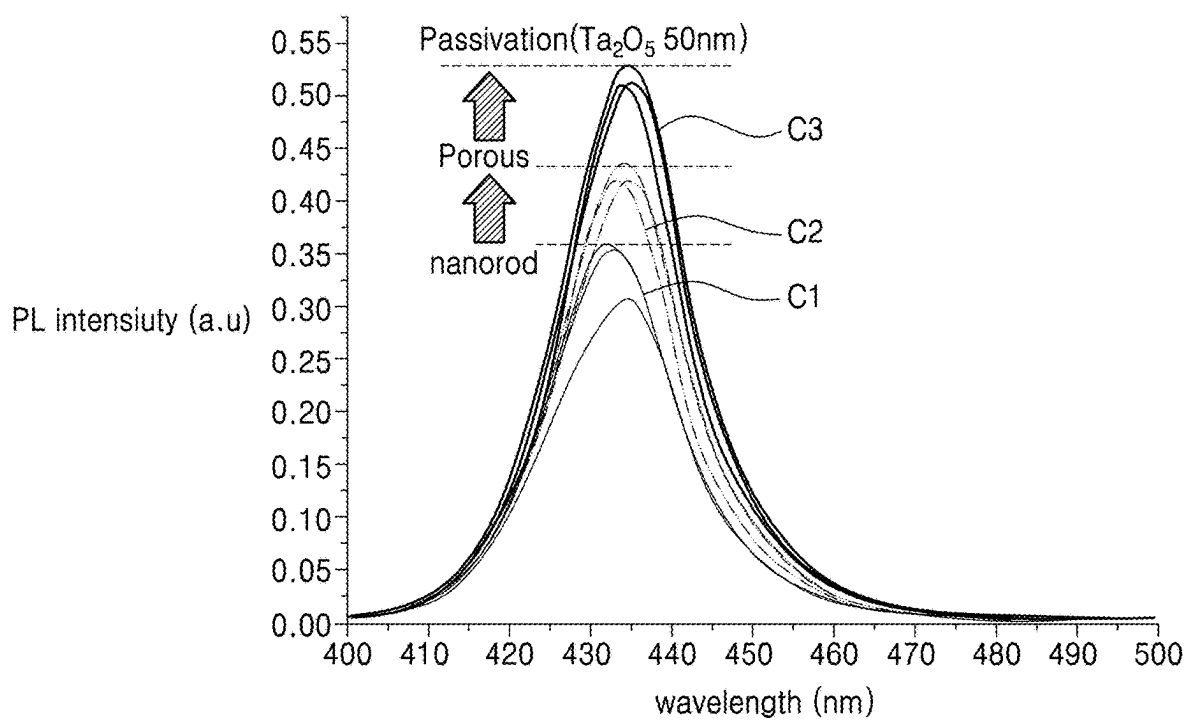


FIG. 4

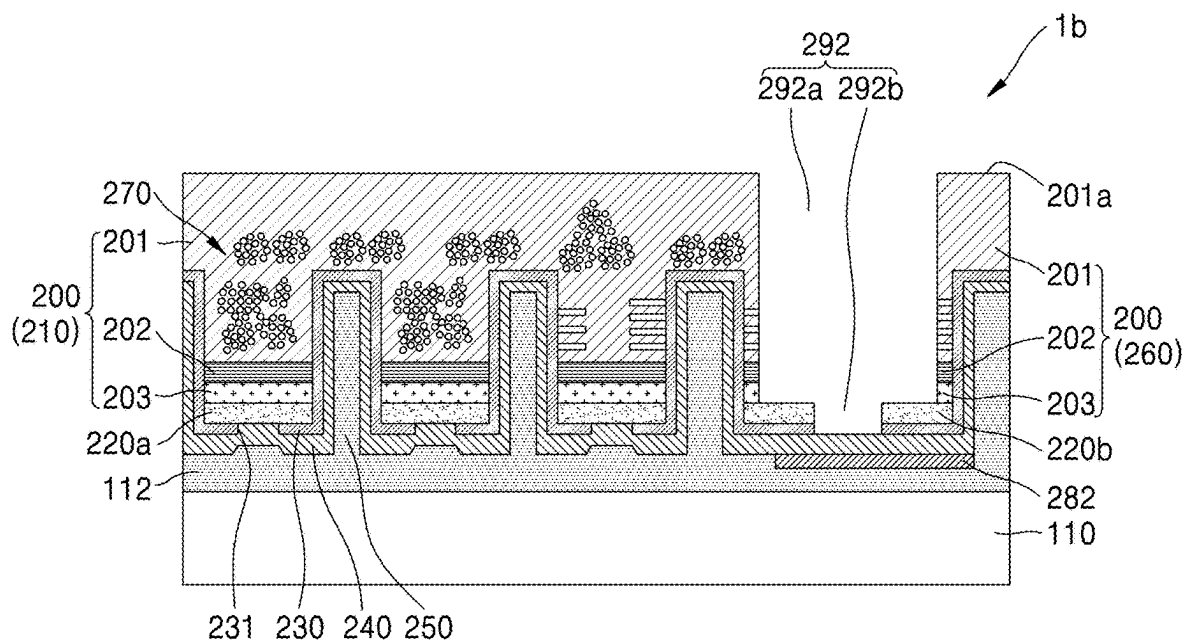


FIG. 5

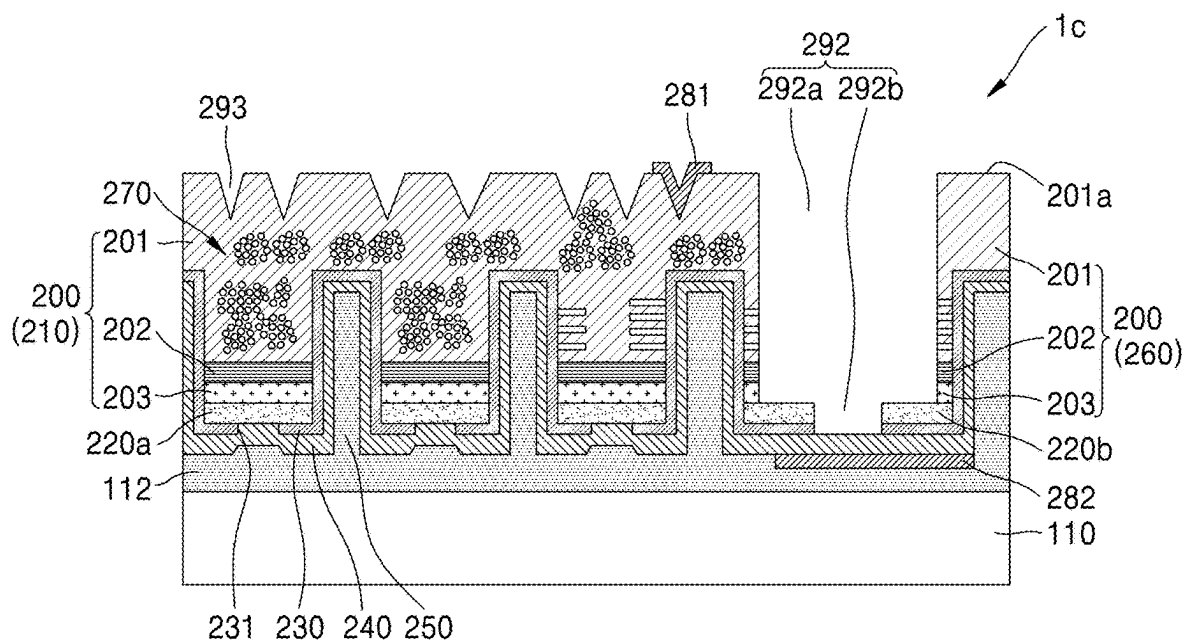


FIG. 6

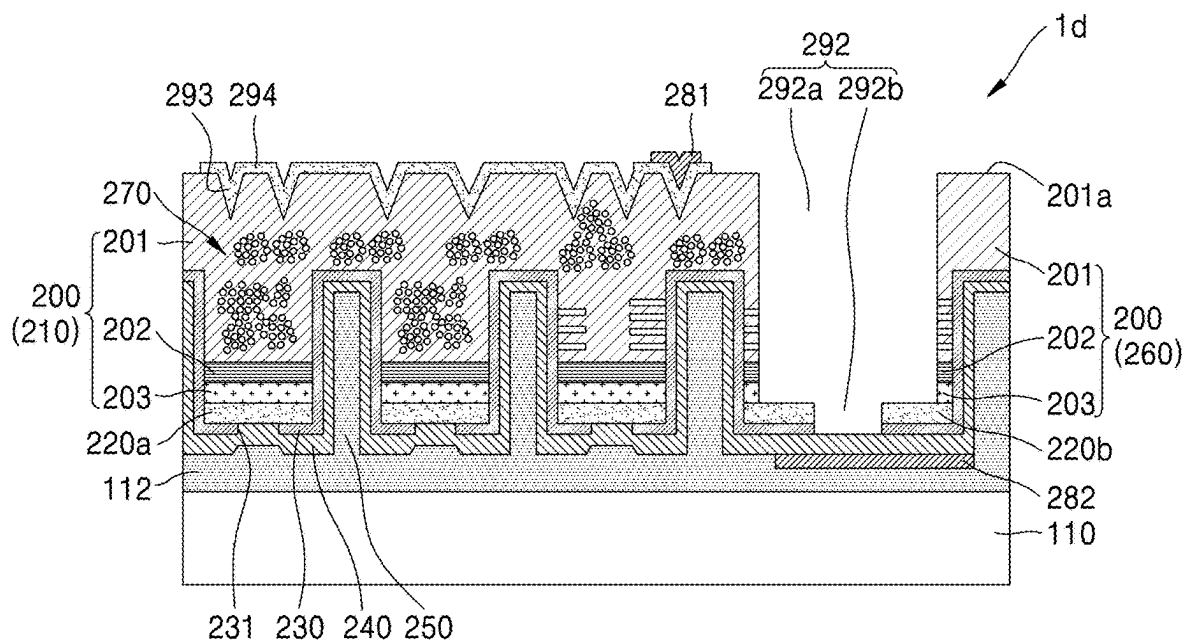


FIG. 7

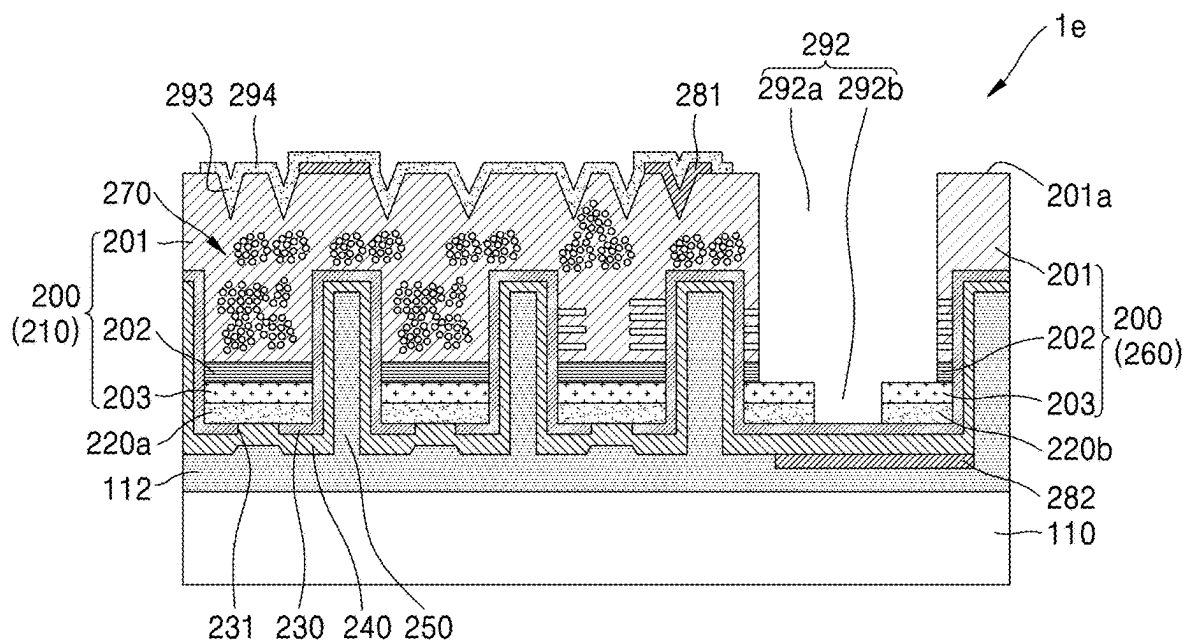


FIG. 8A

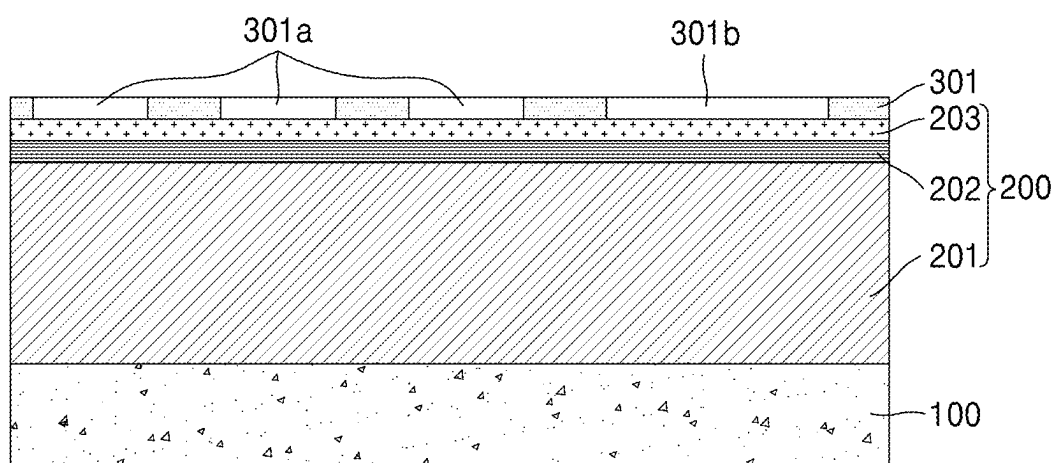


FIG. 8B

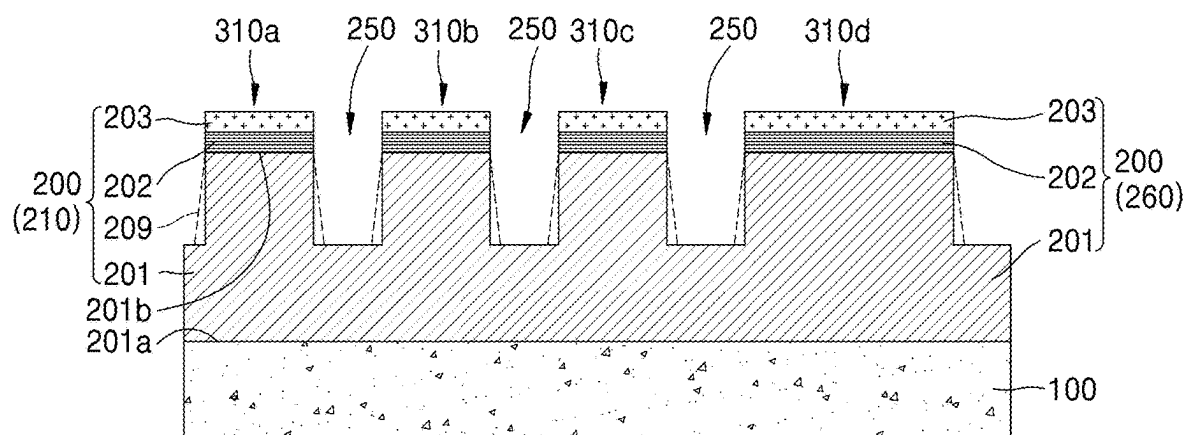


FIG. 8C

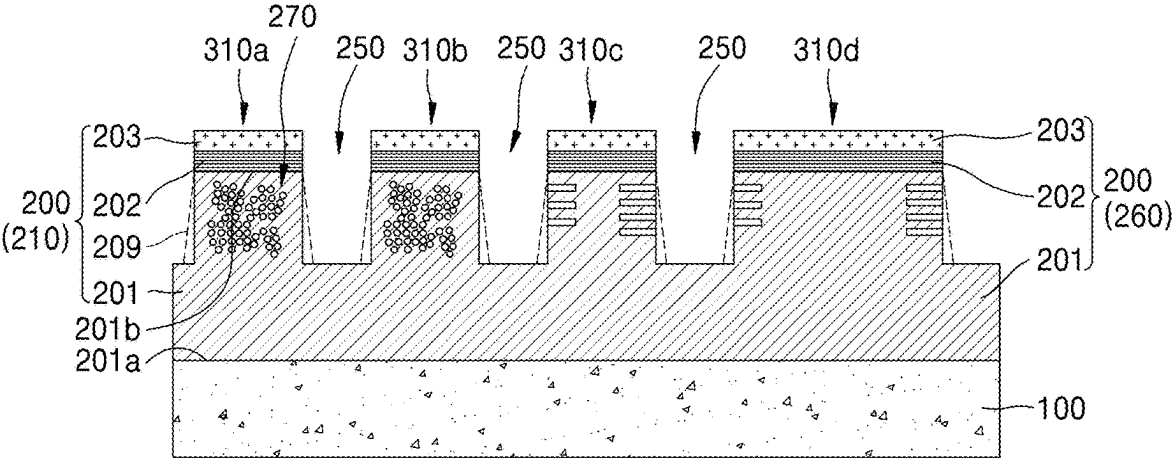


FIG. 8D

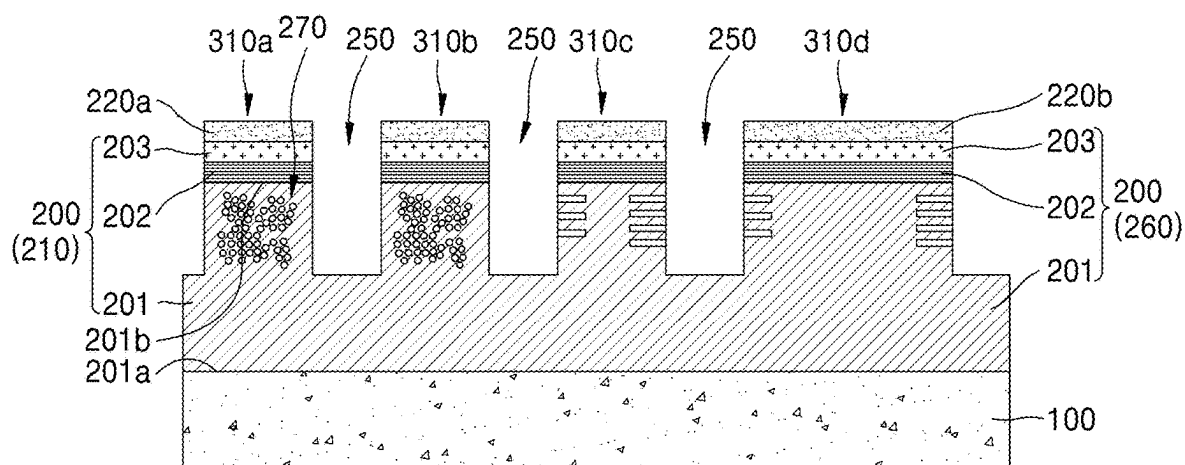


FIG. 8E

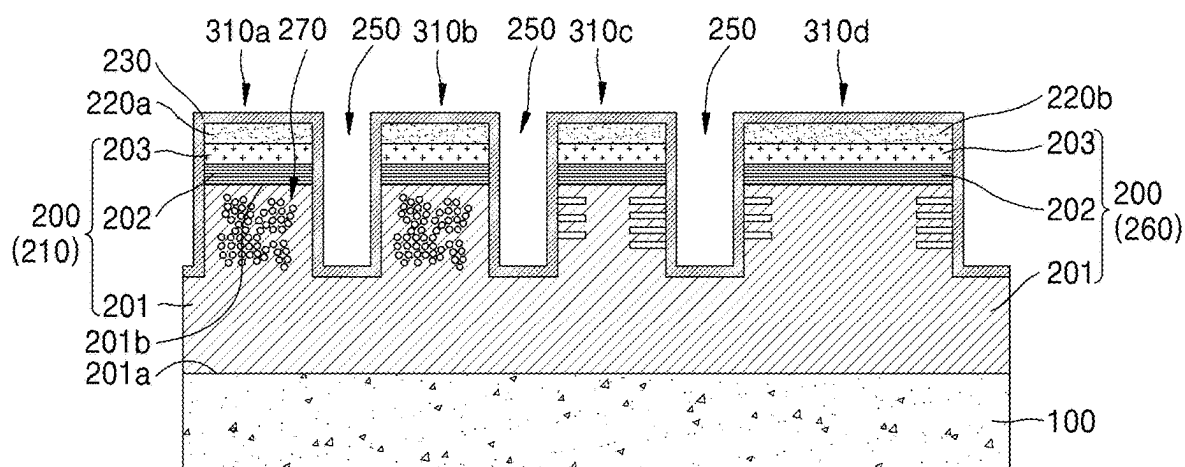


FIG. 8H

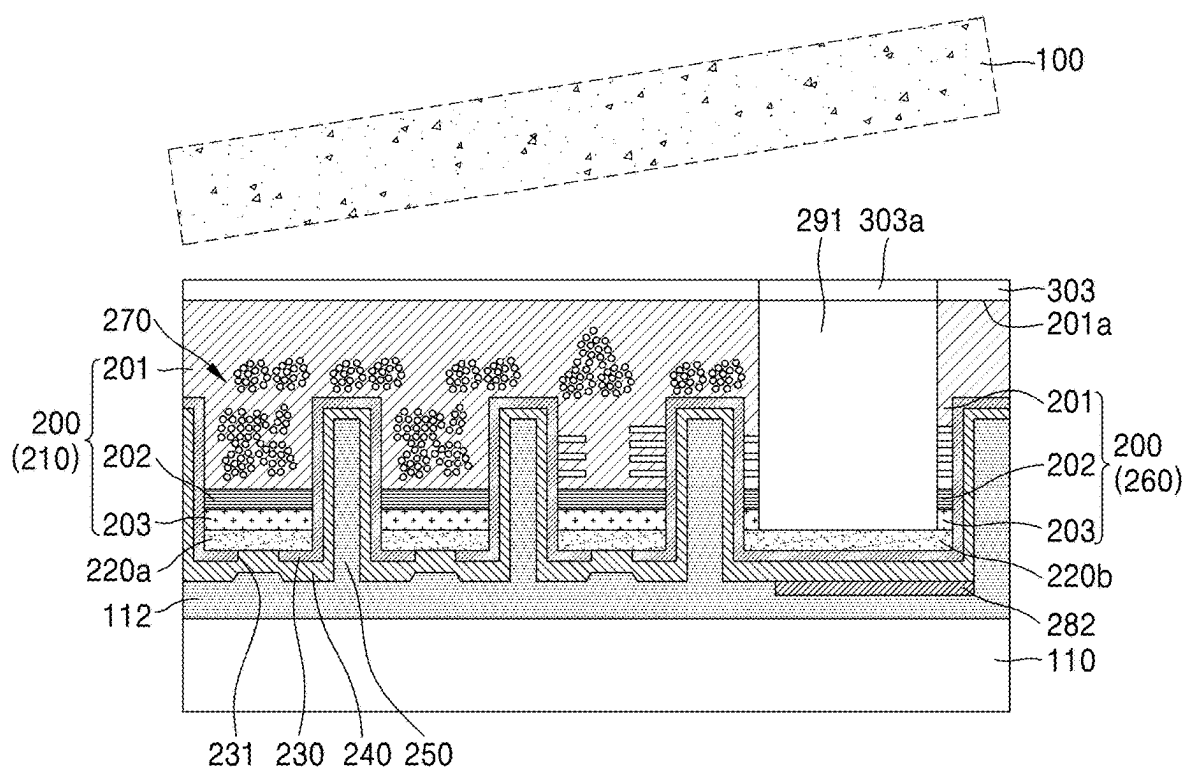


FIG. 8I

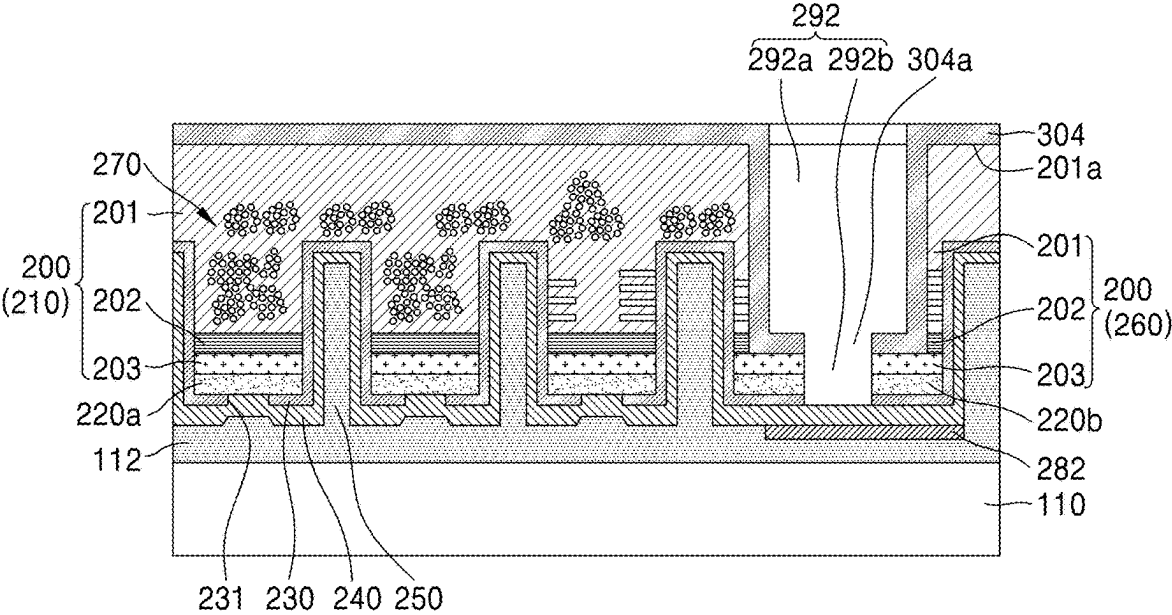


FIG. 9A

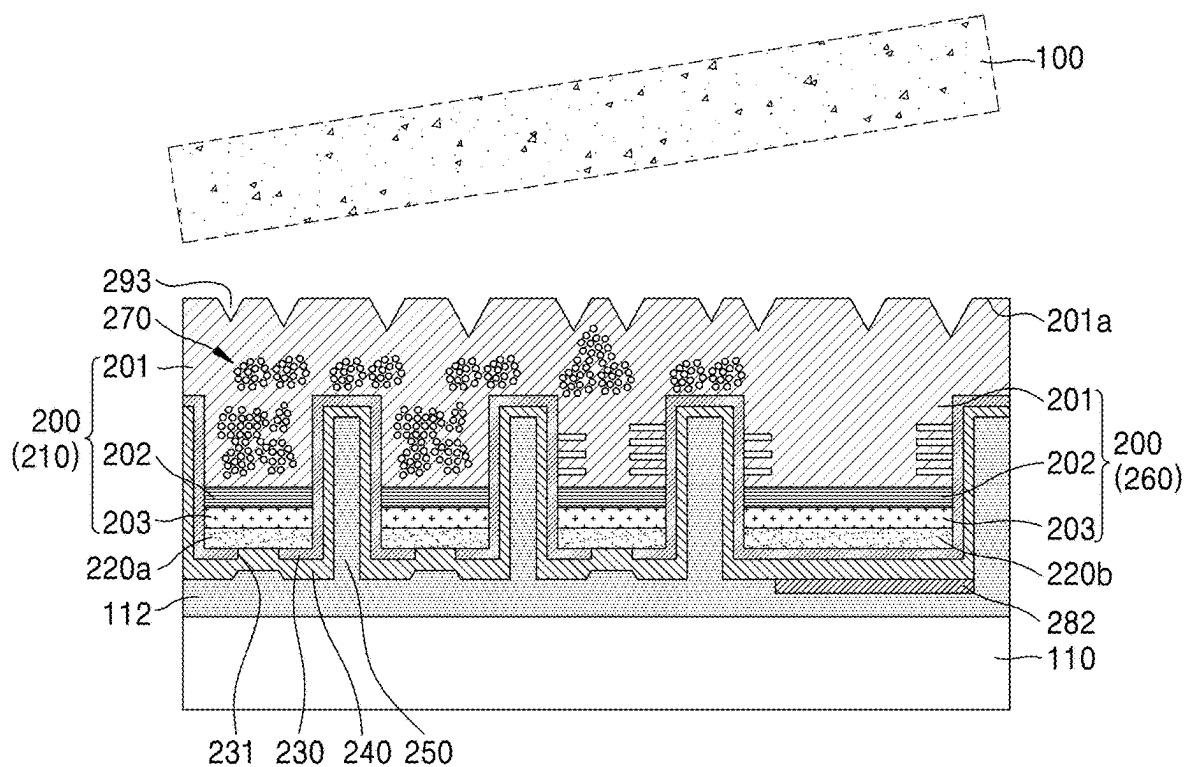


FIG. 9B

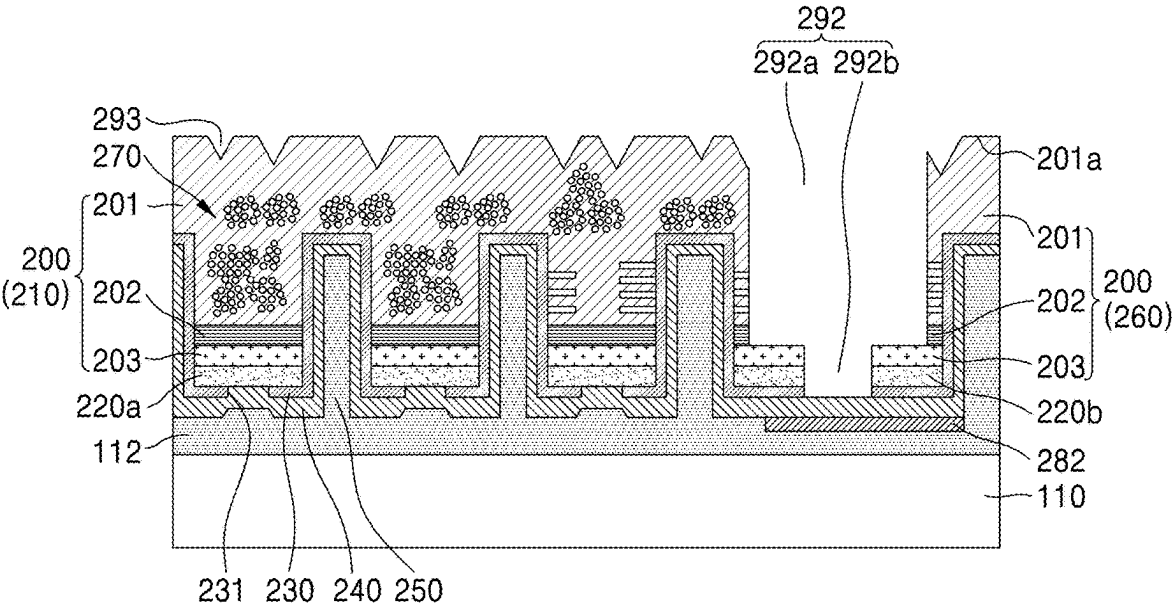
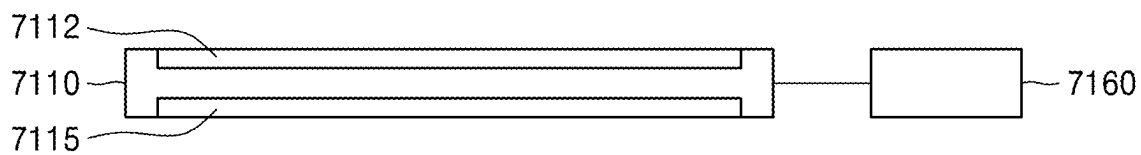


FIG. 10



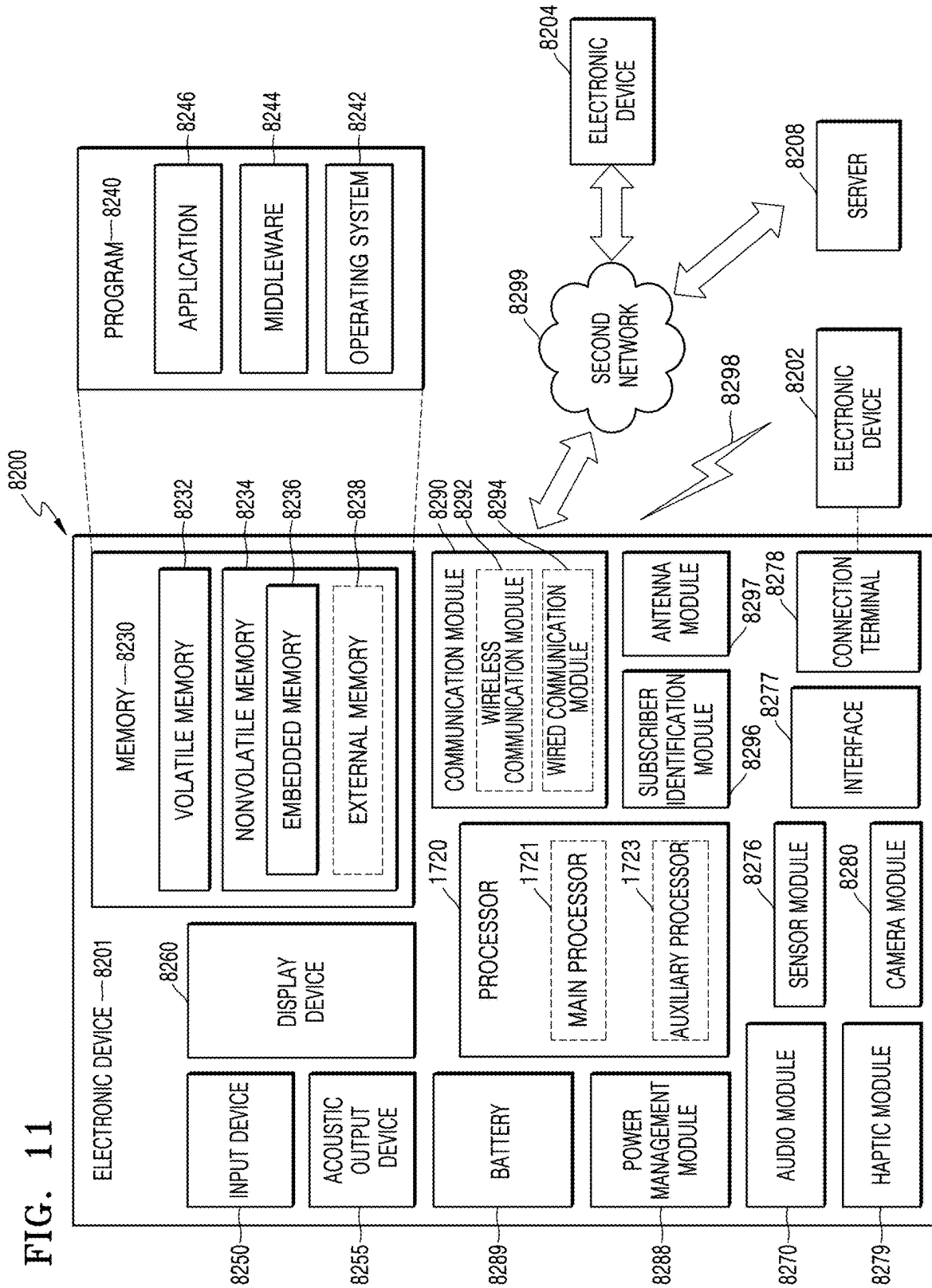


FIG. 12

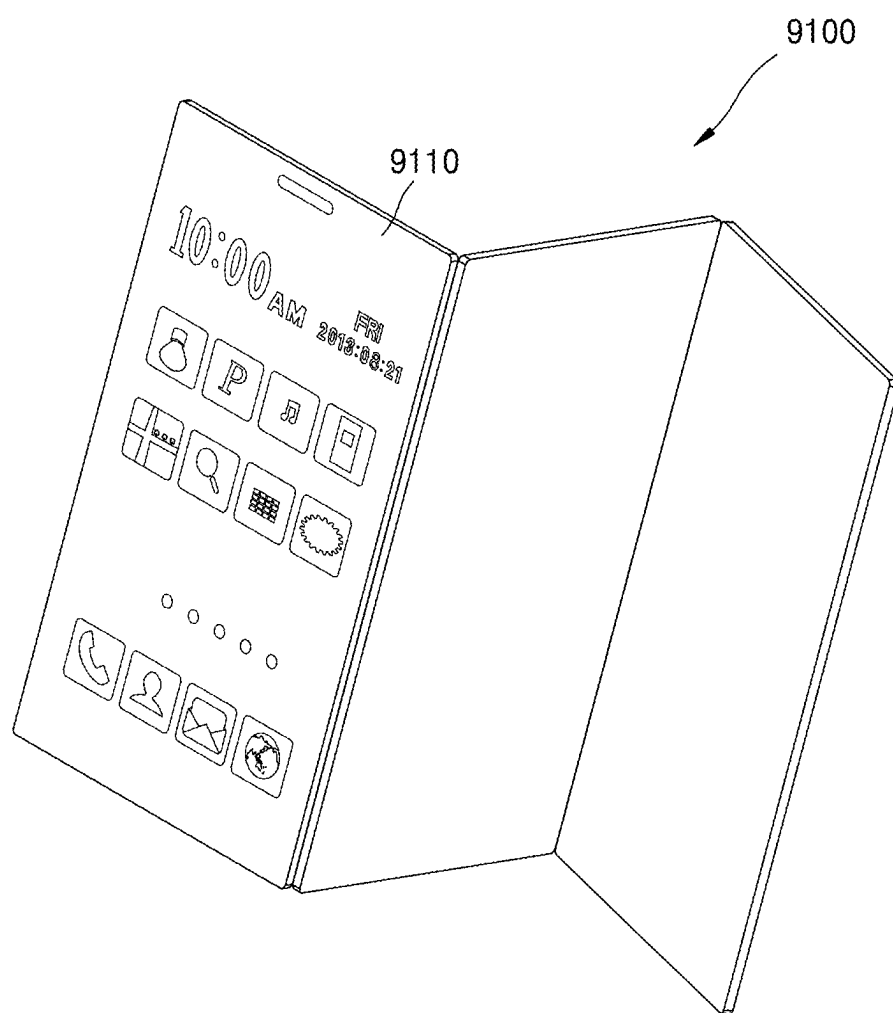


FIG. 13

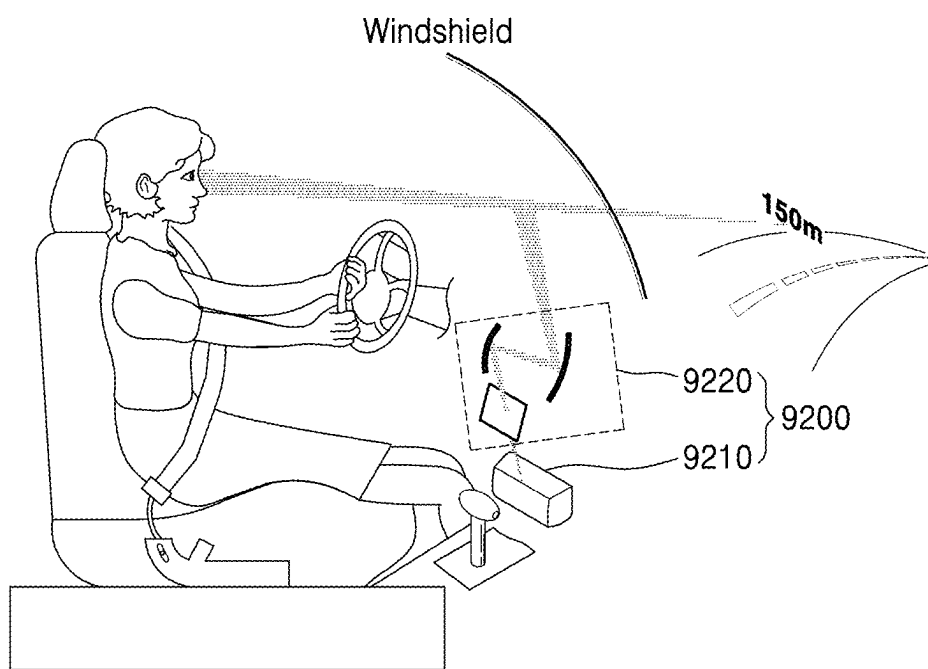


FIG. 14

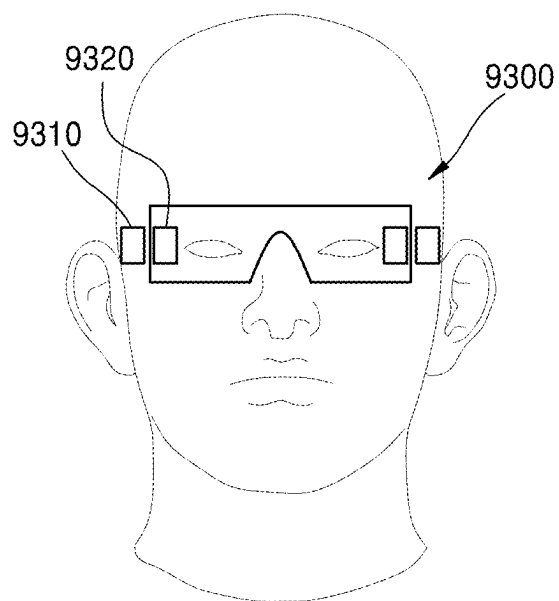


FIG. 15

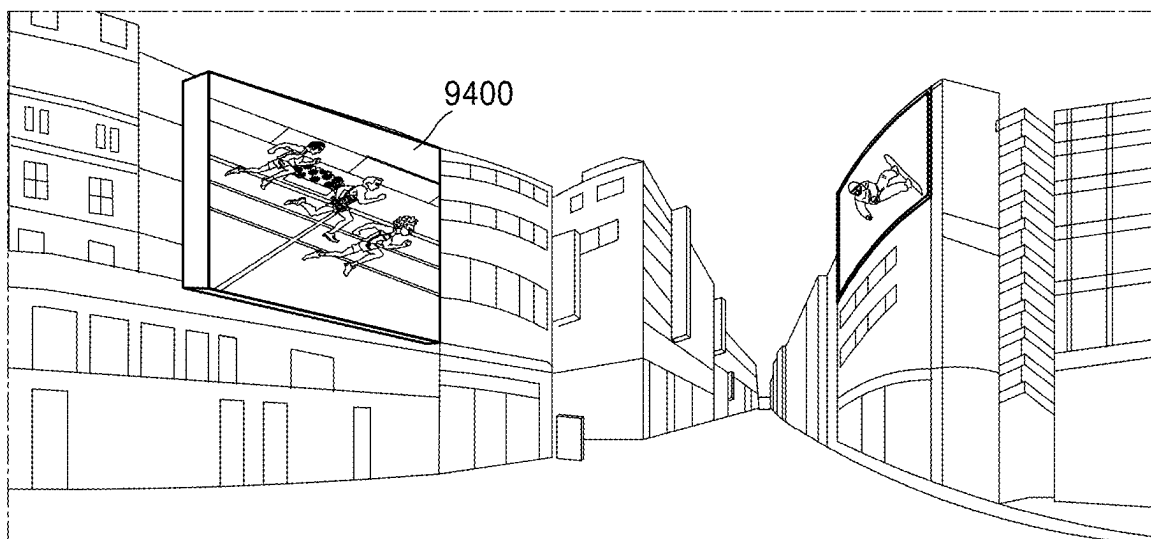
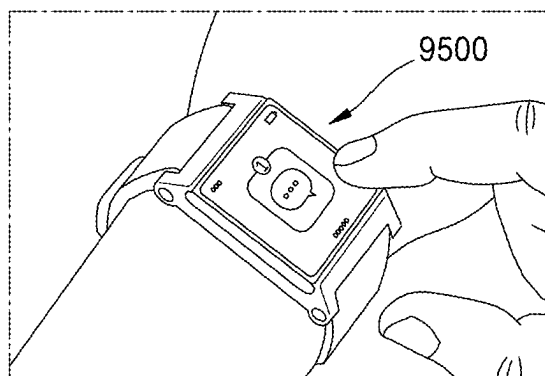


FIG. 16



**LIGHT EMITTING DEVICE, DISPLAY
APPARATUS INCLUDING THE SAME, AND
METHOD OF MANUFACTURING LIGHT
EMITTING DEVICE**

**CROSS-REFERENCE TO RELATED
APPLICATION**

[0001] This application is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2024-0033326, filed on Mar. 8, 2024, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

[0002] The disclosure relates to a light emitting device, a display apparatus including the light emitting device, and a method of manufacturing the light emitting device.

2. Description of the Related Art

[0003] Light emitting diodes (LEDs) are known as the next-generation light sources due to their advantages, such as long lifespan, low power consumption, fast response speed, and environmental friendliness, compared to related art light sources. Because of such advantages, the industrial demand for LEDs is increasing. LEDs have been generally applied and used in various products, such as lighting devices and backlights of display apparatuses.

[0004] Recently, micro units or nano units of micro LEDs using group II-VI or group III-V compound semiconductors have been developed. In addition, micro LED displays including micro LEDs directly used as light emitting devices of display pixels have been developed. However, when LEDs are miniaturized to micro or nano units as described above, the luminous efficiency of the LEDs may be reduced.

SUMMARY

[0005] Provided are a light emitting device with improved luminous efficiency, a method of manufacturing the light emitting device, and a display apparatus including the light emitting device.

[0006] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments of the disclosure. According to an aspect of the disclosure, there is provided a light emitting device including: a light emitting portion and an electrode portion, each of the light emitting portion and the electrode portion including: an epi structure including: a first conductive semiconductor layer including a plurality of pores, an active layer, and a second conductive semiconductor layer, a trench between the light emitting portion, and the electrode portion to expose the first conductive semiconductor layer; a first contact layer on the second conductive semiconductor layer of the light emitting portion; a second contact layer on the second conductive semiconductor layer of the electrode portion; a passivation layer provided on one or more side surfaces of the light emitting portion, and the electrode portion, and on the first contact layer and the second contact layer, the passivation layer including an opening exposing the first contact layer of the light emitting portion; and a

conductive reflective layer provided on the passivation layer and contacting the first contact layer of the light emitting portion through the opening.

[0007] The plurality of pores may be spaced apart from the active layer by at least 100 nm.

[0008] The light emitting device of may further include an electrode pad provided on the conductive reflective layer at a position corresponding to the electrode portion.

[0009] The second contact layer may include Indium Tin Oxide (ITO).

[0010] A via hole penetrating the epi structure may be formed on the electrode portion to expose the second contact layer.

[0011] The via hole may penetrate the second contact layer and the passivation layer to expose the conductive reflective layer.

[0012] The light emitting device may further include an electrode pad provided on a surface of the first conductive semiconductor layer opposite to the active layer.

[0013] The light emitting device may further include a transmission conductive layer provided on a surface of the first conductive semiconductor layer opposite to the active layer.

[0014] The transmission conductive layer may include Indium Tin Oxide (ITO).

[0015] The light emitting device may further include a scattering pattern provided on a surface of the first conductive semiconductor layer opposite to the active layer.

[0016] According to another aspect of the disclosure, there is provided a method of manufacturing a light emitting device, the method including: forming an epi structure including a first conductive semiconductor layer, an active layer, and a second conductive semiconductor layer on a growth substrate; separating a light emitting portion and an electrode portion from each other by forming a trench in the epi structure to expose the first conductive semiconductor layer; forming a first contact layer on the second conductive semiconductor layer of the light emitting portion, and forming a second contact layer on the second conductive semiconductor layer of the electrode portion; forming a plurality of pores in the first conductive semiconductor layer; forming a passivation layer on one or more side surfaces of the light emitting portion and the electrode portion, and on the first contact layer and the second contact layer; forming an opening in the passivation layer to expose the first contact layer of the light emitting portion; and forming a conductive reflective layer covering the passivation layer and contacting the first contact layer of the light emitting portion through the opening.

[0017] The plurality of pores may be spaced apart from the active layer by at least 100 nm.

[0018] The method may further include forming an electrode pad on the conductive reflective layer at a position corresponding to the electrode portion.

[0019] The first contact layer and the second contact layer each may include Indium Tin Oxide (ITO).

[0020] The method may further include transferring the epi structure onto a carrier substrate; and forming a via hole penetrating the first conductive semiconductor layer, the active layer, and the second conductive semiconductor layer of the electrode portion by using the second contact layer as an etch stop layer.

[0021] The via hole may penetrate the second contact layer and the passivation layer to expose the conductive reflective layer.

[0022] The method may further include transferring the epi structure onto a carrier substrate; and forming an electrode pad on a surface of the first conductive semiconductor layer.

[0023] The method may further include forming a scattering pattern on the surface of the first conductive semiconductor layer.

[0024] The method may further include forming a transmission conductive layer on the surface of the first conductive semiconductor layer.

[0025] According to another aspect of the disclosure, there is provided a display apparatus including: a display panel including a plurality of light emitting devices, and a driving circuit configured to switch the plurality of light emitting devices on or off; and a controller configured to input switching signals of the plurality of light emitting devices to the driving circuit according to an image signal, wherein each of the plurality of light emitting devices includes: a light emitting portion and an electrode portion, each of the light emitting portion and the electrode portion including: an epi structure including: a first conductive semiconductor layer including a plurality of pores, an active layer, and a second conductive semiconductor layer, a trench between the light emitting portion, and the electrode portion; a first contact layer on the second conductive semiconductor layer of the light emitting portion; a second contact layer on the second conductive semiconductor layer of the electrode portion; a passivation layer provided on one or more side surfaces of the light emitting portion, and the electrode portion, and on the first contact layer and the second contact layer, the passivation layer including an opening exposing the first contact layer of the light emitting portion; and a conductive reflective layer provided on the passivation layer and contacting the first contact layer of the light emitting portion through the opening.

BRIEF DESCRIPTION OF DRAWINGS

[0026] The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0027] FIG. 1 is a schematic cross-sectional view of a structure of a light emitting device according to an embodiment;

[0028] FIG. 2 is a graph showing a result of simulating photoluminescence (PL) characteristics of a light emitting device;

[0029] FIG. 3 is a schematic cross-sectional view of a light emitting device according to an embodiment;

[0030] FIG. 4 is a schematic cross-sectional view of a light emitting device according to an embodiment;

[0031] FIG. 5 is a schematic cross-sectional view of a light emitting device according to an embodiment;

[0032] FIG. 6 is a schematic cross-sectional view of a light emitting device according to an embodiment;

[0033] FIG. 7 is a schematic cross-sectional view of a light emitting device according to an embodiment;

[0034] FIGS. 8A to 8I are diagrams showing a method of manufacturing a light emitting device according to an embodiment;

[0035] FIGS. 9A and 9B are diagrams showing an operation of forming a scattering pattern, as an embodiment of a method of manufacturing a light emitting device;

[0036] FIG. 10 is a schematic diagram of a display apparatus according to an embodiment;

[0037] FIG. 11 is a block diagram of an electronic device including a display according to an embodiment;

[0038] FIG. 12 illustrates an embodiment of a mobile device as an application example of an electronic device;

[0039] FIG. 13 illustrates an embodiment of a vehicle head-up display apparatus as an application example of an electronic device;

[0040] FIG. 14 illustrates an embodiment of augmented reality glasses or virtual reality glasses as an application example of an electronic device;

[0041] FIG. 15 illustrates an embodiment of a signage as an application example of an electronic device; and

[0042] FIG. 16 illustrates an embodiment of a wearable display as an application example of an electronic device.

DETAILED DESCRIPTION

[0043] Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout. In this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described below, by referring to the figures, to explain aspects. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

[0044] Hereinafter, a display apparatus and a method of manufacturing the display apparatus will be described in detail with reference to the accompanying drawings. In the following drawings, the same reference numerals refer to the same components, and the size of each component in the drawings may be exaggerated for clarity and convenience of description. The embodiments described below are merely exemplary, and various modifications are possible from these embodiments.

[0045] In the following description, when a component is referred to as being “above” or “on” another component, it may be directly above or on the other component while making contact with the other component or may be above or on the other component without making contact with the other component. The terms of a singular form may include plural forms unless otherwise specified. In addition, when a certain part “includes” a certain component, it means that other components may be further included rather than excluding other components unless otherwise stated.

[0046] The use of the term “the” and similar designating terms may correspond to both the singular and the plural. When there is no explicit description of the order of operations constituting a method or no contrary description thereto, these operations may be performed in an appropriate order, and are not limited to the order described.

[0047] The connecting lines, or connectors shown in the various figures presented are intended to represent functional relationships and/or physical or logical couplings between the various elements. It should be noted that many

alternative or additional functional relationships, physical connections or logical connections may be present in a practical device.

[0048] The use of all illustrative terms is merely for describing technical ideas in detail, and the scope is not limited by these examples or illustrative terms unless limited by the claims.

[0049] FIG. 1 is a schematic cross-sectional view of a structure of a light emitting device 1 according to an embodiment. The light emitting device 1 of the embodiment is a flip-chip type light emitting device. However, the disclosure is not limited thereto, and as such, the light emitting device 1 may be of a different type. In FIG. 1, for convenience, the light emitting device 1 before being flipped is shown.

[0050] Referring to FIG. 1, the light emitting device 1 may include a light emitting portion 210 and an electrode portion 260. The light emitting portion 210 and the electrode portion 260 may each include an epi structure 200. The epi structure 200 may include a first conductive semiconductor layer 201, an active layer 202, and a second conductive semiconductor layer 203. For example, the first conductive semiconductor layer 201, the active layer 202, and the second conductive semiconductor layer 203 are sequentially stacked. The light emitting portion 210 and the electrode portion 260 are separated from each other by a trench 250. A first contact layer 220a and a second contact layer 220b may be provided on the second conductive semiconductor layers 203 of the light emitting portion 210 and the electrode portion 260, respectively. For example, in the light emitting portion 210, the first contact layer 220a may be provided on the second conductive semiconductor layers 203 of the light emitting portion 210, and in the electrode portion 260, the second contact layer 220b may be provided on the second conductive semiconductor layers 203 of the electrode portion 260. A passivation layer 230 may be provided on one or side surfaces of the light emitting portion 210 and the electrode portion 260, and the first contact layer 220a and the second contact layer 220b. For example, the passivation layer 230 may cover the side surfaces of the light emitting portion 210 and the electrode portion 260, and the first contact layer 220a and the second contact layer 220b. An opening 231 is provided in the passivation layer 230 to expose the first contact layer 220a of the light emitting portion 210. According to an embodiment, the conductive reflective layer 240 may be provided on the passivation layer 230. For example, the conductive reflective layer 240 may cover the passivation layer 230. The conductive reflective layer 240 contacts the first contact layer 220a of the light emitting portion 210 through the opening 231. A plurality of pores 270 may be provided in the first conductive semiconductor layer 201.

[0051] The light emitting portion 210 may be referred to as a nanorod light emitting portion. The light emitting device 1 may include a plurality of light emitting portions 210 in the form of nanorods. The plurality of light emitting portions 210 are separated from each other by the trenches 250. The electrode portion 260 may entirely correspond to the plurality of light emitting portions 210. The light emitting device 1 may correspond to one pixel or sub-pixel in a display apparatus.

[0052] For example, the epi structure 200 may be formed on a substrate 100. The substrate 100 may be referred to as a growth substrate. The substrate 100 is, for example, a growth substrate for semiconductor single crystal growth,

and may use a silicon (Si) substrate, a silicon carbide (SiC) substrate, a sapphire substrate, etc. Besides the above, a substrate including a material, suitable for the growth of the epi structure 200 to be formed on the substrate 100, for example, AlN, AlGaIn, ZnO, GaAs, MgAl₂O₄, MgO, LiAlO₂, LiGaO₂, GaN, may be used. According to an apparatus to which the light emitting device 1 is applied, the substrate 100 may be removed after completely manufacturing the light emitting device 1. According to an embodiment, a buffer layer for epitaxial growth of the epi structure 200 may be provided on a surface of the substrate 100, and the epi structure 200 may be grown on the buffer layer.

[0053] The epi structure 200 may include a Group III-V nitride semiconductor material. The Group III-V nitride semiconductor material may include, for example, GaN, InGaIn, AlInGaIn, AlGaInP, etc. For example, the epi structure 200 may include a GaN-based semiconductor material. The epi structure 200 has a structure in which the first conductive semiconductor layer 201, the active layer 202 having a quantum well structure, and the second conductive semiconductor layer 203 are sequentially stacked.

[0054] The first conductive semiconductor layer 201 may be formed on the substrate 100 by growing in a direction perpendicular to the surface of the substrate 100. The first conductive semiconductor layer 201 may be a semiconductor layer doped with first type impurities. For example, the first conductive semiconductor layer 201 may be an n-GaN layer doped with n-type impurities. Si, Ge, Se, Te, etc. may be used as n-type impurities.

[0055] The active layer 202 is a layer that emits light through electron-hole recombination. The active layer 202 may be formed by growing on the first conductive semiconductor layer 201. The active layer 202 has the quantum well structure. For example, the active layer 202 may have a single quantum well structure or a multi quantum well structure obtained by periodically changing x, y, and z values in Al_xGa_yIn_z and adjusting a band gap. For example, a quantum well layer and a barrier layer may be paired in the form of InGaIn/GaN, InGaIn/InGaIn, InGaIn/AlGaIn, or InGaIn/InAlGaIn to form the quantum well structure, and a band gap energy may be controlled according to a composition ratio of indium (In) in a material layer including indium (In), and an emission wavelength band may be adjusted.

[0056] The second conductive semiconductor layer 203 may be formed on the active layer 202. The second conductive semiconductor layer 203 may be formed by growing on the active layer 202. The second conductive semiconductor layer 203 may be a semiconductor layer doped with second type impurities. For example, the second conductive semiconductor layer 203 may be a p-GaN layer doped with p-type impurities. Mg, Zn, Be, etc. may be used as p-type impurities.

[0057] The epi structure 200 may be formed using hybrid vapor phase epitaxy (HVPE), molecular beam epitaxy (MBE), metal organic vapor phase epitaxy (MOVPE), metal organic chemical vapor deposition (MOCVD), etc.

[0058] The light emitting portion 210 and the electrode portion 260 are separated from each other by the trench 250 formed by immersing in the epi structure 200. The trench 250 is formed by passing through the second conductive semiconductor layer 203 and the active layer 202 and immersing to the first conductive semiconductor layer 201. The trench 250 is formed by immersing in an arbitrary

position between two surfaces of the first conductive semiconductor layer 201. The light emitting portion 210 and the electrode portion 260 share the first conductive semiconductor layer 201.

[0059] In FIG. 1, the side surfaces of the light emitting portion 210 and the electrode portion 260 are shown perpendicular to the substrate 100, but the disclosure is not limited thereto. As shown by a dashed line 209 in FIG. 1, the side surfaces of the light emitting portion 210 and the electrode portion 260 may have a tapered shape with respect to the substrate 100. For example, the first conductive semiconductor layer 201 may have the tapered shape from a surface 201b on the active layer 202 toward a surface 201a opposite to the active layer 202. This is due to the difference in an etch rate between near the surface 201b of the first conductive semiconductor layer 201 close to an etch mask in a process of etching the epi structure 200 and near the surface 201a of the first conductive semiconductor layer 201. A width of the first conductive semiconductor layer 201 may gradually increase from the surface 201b to the surface 201a. In other words, a width of the trench 250 may gradually decrease from the surface 201b toward the surface 201a.

[0060] The first contact layer 220a is formed on the second conductive semiconductor layer 203 of the light emitting portion 210. The first contact layer 220a forms an ohmic contact with the second conductive semiconductor layer 203. The first contact layer 220a may include a conductive material. The conductive material may include Al, Ti, Pt, Ag, Au, Pd, TiW, or various combinations thereof. The first contact layer 220a may include a transparent conductive material. The transparent conductive material may include ITO (Indium Tin Oxide). The second contact layer 220b is formed on the second conductive semiconductor layer 203 of the electrode portion 260. The second contact layer 220b may include a conductive material. The conductive material may include Al, Ti, Pt, Ag, Au, Pd, TiW, or various combinations thereof. The second contact layer 220b may include a transparent conductive material. The transparent conductive material may include ITO.

[0061] According to an embodiment, the second contact layer 220b may include a material that may function as an etch stop layer with respect to a semiconductor material forming the epi structure 200, for example, GaN. The second contact layer 220b may include, for example, a material having a lower etch rate than that of GaN in an etching process. For example, the second contact layer 220b may include ITO. According to experiments, the etch rate of GaN is about 8.2 nm/s, which is about 19.5 times the etch rate of ITO, which is about 0.42 nm/s. Mathematically, an ITO layer with a thickness of about 100 nm may correspond to a GaN layer with a thickness of about 2 μm in terms of etch thickness, and the second contact layer 220b including ITO may function as an excellent etch stop layer. Accordingly, as is described below, in an example case in which the light emitting device 1 is flipped and used, damage to the conductive reflective layer 240 may be prevented in a process of etching GaN to form a via hole (FIG. 3:291) exposing the conductive reflective layer 240, which is a p-electrode, to the electrode portion 260.

[0062] According to an embodiment, the first contact layer 220a and the second contact layer 220b may include the same material. For example, the first contact layer 220a and the second contact layer 220b may include ITO. According

to such a configuration, the first contact layer 220a and the second contact layer 220b may be formed through a single process, and thus, a manufacturing process of the light emitting device 1 may be simplified.

[0063] According to an embodiment, the passivation layer 230 may be provided on the side surfaces of the light emitting portion 210 and the electrode portion 260. The passivation layer 230 may be provided on the first contact layer 220a and the second contact layer 220b. The passivation layer 230 may be provided on the side surfaces of the first conductive semiconductor layer 201, the active layer 202, and the second conductive semiconductor layer 203 exposed by the trench 250, an upper surface of the first conductive semiconductor layer 201 exposed by the trench 250, and the side surfaces and upper surfaces of the first contact layer 220a and the second contact layer 220b. For example, the passivation layer 230 covers the side surfaces of the light emitting portion 210 and the electrode portion 260. The passivation layer 230 covers the first contact layer 220a and the second contact layer 220b. The passivation layer 230 covers the side surfaces of the first conductive semiconductor layer 201, the active layer 202, and the second conductive semiconductor layer 203 exposed by the trench 250, an upper surface of the first conductive semiconductor layer 201 exposed by the trench 250, and the side surfaces and upper surfaces of the first contact layer 220a and the second contact layer 220b. The passivation layer 230 may protect the light emitting portion 210 from external physical and chemical impacts. The passivation layer 230 may prevent current leakage by insulating the light emitting portion 210. The opening 231 is provided in the passivation layer 230. The opening 231 partially exposes, for example, the upper surface of the first contact layer 220a of the light emitting portion 210. The passivation layer 230 may have a single-layer structure or a multi-layer structure. The passivation layer 230 may include a dielectric material. The dielectric material may include SiO_2 , TiO_2 , Si_3N_4 , AlO_x , AlO_xN_y , Ta_2O_5 , TiN, AlN, ZrO_2 , TiAlN, TiSiN, HfO_x , or various combinations thereof. A thickness of the passivation layer 230 may be about 5 nm or more and about 50 nm or less, for example, about 10 nm or more and about 30 nm or less. According to an embodiment, in an example case in which the light emitting portion 210 emits red light, the passivation layer 230 may include AlO_xN_y , and accordingly, the luminous efficiency of red light may be improved. According to an embodiment, in an example case in which the light emitting portion 210 emits blue light, the passivation layer 230 may include Ta_2O_5 . As a result, the luminous efficiency of blue light may be improved.

[0064] The conductive reflective layer 240 may be provided the passivation layer 230. The conductive reflective layer 240 may include a conductive material. The conductive reflective layer 240 may include a conductive material having reflectivity. For example, the conductive material may include Al, Ti, Pt, Ag, Au, Pd, TiW, or various combinations thereof. According to an embodiment, the conductive reflective layer 240 may be provided on an entire area of the passivation layer 230. For example, the conductive reflective layer 240 may be provided on areas of the passivation layer 230 corresponding to the light emitting portion 210 and the electrode portion 260. For example, the conductive reflective layer 240 may cover the entire area of the passivation layer 230 including areas of the passivation layer 230 corresponding to the light emitting portion 210 and the

electrode portion 260. The conductive reflective layer 240 is in contact with the first contact layer 220a of the light emitting portion 210 through the opening 231. Because the passivation layer 230 is disposed between the conductive reflective layer 240 and the second contact layer 220b of the electrode portion 260, the conductive reflective layer 240 does not contact the second contact layer 220b of the electrode portion 260. The conductive reflective layer 240 functions as an electrode of the plurality of light emitting portions 210, that is, as a p-electrode of the plurality of light emitting portions 210 in the embodiment. The conductive reflective layer 240 may be connected to a driving circuit of the display apparatus. For example, the conductive reflective layer 240 may be connected to a drain of a driving transistor.

[0065] An electrode pad (second electrode pad) 282 may be further provided on the conductive reflective layer 240. The second electrode pad 282 may be provided at a position corresponding to the electrode portion 260 on the conductive reflective layer 240. The second electrode pad 282 may include a conductive material. For example, the conductive material may include Al, Ti, Pt, Ag, Au, Pd, TiW, or various combinations thereof. In the embodiment, the second electrode pad 282 functions as a p-pad. The electrode pad 282 may be connected to the driving circuit of the display apparatus, for example, the drain of a driving transistor.

[0066] According to an embodiment, the light emitting device 1 may further include an electrode pad (first electrode pad) electrically connected to the first conductive semiconductor layer 201. The first electrode pad may be connected to a power line in the driving circuit of the display apparatus. In the display apparatus including the plurality of light emitting devices 1, the first electrode pad may be a common electrode.

[0067] The light emitting portion 210 includes the plurality of pores 270. The plurality of pores 270 are provided in the first conductive semiconductor layer 201. The plurality of pores 270 may be formed to extend in a transverse direction inward from the side surfaces of the first conductive semiconductor layer 201. In order to show shapes of the pore 270 viewed from various directions, the pores 270 are shown in FIG. 1 in a circular shape and a shape extending in the transverse direction. Among lights emitted from the active layer 202, light having an incident angle with respect to the surface 201a of the first conductive semiconductor layer 201 being more than a critical angle is emitted to the outside through the surface 201a of the first conductive semiconductor layer 201, and the remaining light is totally reflected on the surface 201a of the first conductive semiconductor layer 201. As described above, light trapped inside the light emitting device 1 by total internal reflection may be a factor that reduces the light extraction efficiency of the light emitting device 1. The plurality of pores 270 act as scattering centers of light emitted from the active layer 202. The totally reflected light is scattered by the plurality of pores 270 such that a traveling direction thereof changes, is incident again on the surface 201a of the first conductive semiconductor layer 201, and the light having the incident angle with respect to the surface 201a of the first conductive semiconductor layer 201 being more than the critical angle is emitted to the outside through the surface 201a of the first conductive semiconductor layer 201. As described above, the plurality of pores 270 scatter the light trapped inside the light emitting device 1 by total internal reflection, thereby changing the traveling direction of the light and allowing the

light to be discharged from the light emitting device 1. As a result, the light emitting device 1 may have improved light extraction efficiency.

[0068] Light is generated by electron-hole recombination inside the active layer 202. In an example case in which the pore 270 exists within the active layer 202, a surface that causes surface recombination is provided at a boundary between a material forming the active layer 202 and the pore 270. Because the surface of a material generally has more crystal defects than the inside of the material, surface recombination is easier than inside recombination. Surface recombination is non-emission recombination, and may cause a decrease in the luminous efficiency of the active layer 202. Therefore, it is necessary to prevent the pore 270 from being formed in the active layer 202, and in the embodiment, the plurality of pores 270 may be formed to be spaced at least 100 nm apart from the active layer 202. For example, in FIG. 1, a separation distance 270D between the pore 270 closest to the active layer 202 and the active layer 202 may be 100 nm or more.

[0069] FIG. 2 is a graph showing a result of simulating photoluminescence (PL) characteristics of the light emitting device 1. In FIG. 2, graph C1 represents a case in which the plurality of pores 270 are not applied, graph C2 represents a case in which the plurality of pores 270 are applied, and graph C3 represents a case in which the plurality of pores 270 and the passivation layer 230 including Ta₂O₅ are applied. FIG. 2 shows that the case in which the plurality of pores 270 are applied (graph C2) has higher PL characteristics than the case in which the plurality of pores 270 are not applied (graph C1). Also, FIG. 2 shows that the case in which the passivation layer 230 including Ta₂O₅ is additionally applied has higher PL characteristics than the case in which the plurality of pores 270 are not applied (graph C1) and the case in which the plurality of pores 270 are applied (graph C2).

[0070] FIG. 3 is a schematic cross-sectional view of a light emitting device 1a according to an embodiment. The light emitting device 1a of the embodiment is a flip-chip type light emitting device. FIG. 3 shows that the light emitting device 1a is transferred to a carrier substrate 110 and flipped. The light emitting device 1a is attached to the carrier substrate 110 with an adhesive 112. The light emitting device 1a of the embodiment is different from the light emitting device 1 shown in FIG. 1 in that the electrode portion 260 is disabled (non-emitting). Hereinafter, components that are the same as those of the light emitting device 1 shown in FIG. 1 are denoted by the same reference numerals, and differences from the light emitting device 1 are mainly described. The description of the light emitting device 1 applies to the light emitting device 1a unless contradictory.

[0071] Referring to FIG. 3, a via hole 291 is formed through the epi structure 200 to expose the second contact layer 220b on the electrode portion 260. The via hole 291 penetrates the first conductive semiconductor layer 201, the active layer 202, and the second conductive semiconductor layer 203. The via hole 291 is formed from the surface 201a opposite to the active layer 202 among two surfaces of the first conductive semiconductor layer 201 in a stacking direction to the second contact layer 220b. By the via hole 291, an area of the epi structure 200 corresponding to the

electrode portion 260 is disabled (non-emitting). As a result, the conductive reflective layer 240 that may function as a p-electrode may be exposed.

[0072] When forming the via hole 291 as described above, the second contact layer 220b may function as an etch stop layer. The second contact layer 220b may include a material, for example, ITO, having a large difference in an etch rate from the epi structure 200. ITO has a very low etch rate compared to GaN forming the epi structure 200. According to such a configuration, the risk of damage to the conductive reflective layer 240, which functions as the p-electrode, and the second electrode pad 282 in a process of forming the via hole 291 may be reduced.

[0073] FIG. 4 is a schematic cross-sectional view of a light emitting device 1b according to an embodiment. The light emitting device 1b of this embodiment is a flip-chip type light emitting device. FIG. 4 shows that the light emitting device 1b is transferred and flipped to a carrier substrate 110. The light emitting device 1b is attached to the carrier substrate 110 with the adhesive 112. The light emitting device 1b of the embodiment is different from the light emitting device 1b shown in FIG. 3 in that a via hole 292 extends to the conductive reflective layer 240. Hereinafter, components that are the same as those of the light emitting device 1a shown in FIG. 3 are denoted by the same reference numerals, and differences from the light emitting device 1a are mainly described. Descriptions of the light emitting device 1 and the light emitting device 1a apply to the light emitting device 1b, unless contradictory.

[0074] Referring to FIG. 4, the via hole 292 is provided in the electrode portion 260. The via hole 292 penetrates the epi structure 200, the second contact layer 220b, and the passivation layer 230 to expose the conductive reflective layer 240. The via hole 292 may include a first via hole 292a penetrating the epi structure 200, and a second via hole 292b extending from the first via hole 292a to penetrate the second contact layer 220b and the passivation layer 230. A diameter of the second via hole 292b may be less than that of the first via hole 292a. The first via hole 292a is formed from the surface 201a opposite to the active layer 202 among two surfaces of the first conductive semiconductor layer 201 in a stacking direction to the second contact layer 220b. By the first via hole 292a, an area of the epi structure 200 corresponding to the electrode portion 260 is disabled (non-emitting). The via hole 292 may function as a connection hole for connecting a driving circuit of a display apparatus and a p-electrode. The driving circuit may be connected to the conductive reflective layer 240 through the via hole 292.

[0075] FIG. 5 is a schematic cross-sectional view of a light emitting device 1c according to an embodiment. The light emitting device 1c of the embodiment is a flip-chip type light emitting device. FIG. 5 shows that the light emitting device 1c is transferred and flipped to the carrier substrate 110. The light emitting device 1c is attached to the carrier substrate 110 with the adhesive 112. The light emitting device 1c of the embodiment is different from the light emitting device 1b shown in FIG. 4 in that the light emitting device 1c includes a scattering pattern 293. Hereinafter, components that are the same as those of the light emitting device 1b shown in FIG. 4 are denoted by the same reference numerals, and differences from the light emitting device 1b are mainly described. Descriptions of the light emitting device 1, the

light emitting device 1a, and the light emitting device 1b apply to the light emitting device 1c, unless contradictory.

[0076] Referring to FIG. 5, the light emitting device 1c has the scattering pattern 293. The scattering pattern 293 is provided on the surface 201a of the first conductive semiconductor layer 201 opposite to the active layer 202. The scattering pattern 293 may be formed in an area of the surface 201a of the first conductive semiconductor layer 201 at least corresponding to the light emitting portion 210. The scattering pattern 293 may be, for example, an uneven pattern that may scatter light. In an example case in which light emitted from the active layer 202 is incident on the scattering pattern 293, the light is scattered by the scattering pattern 293. As a result, the light emitted from the light emitting device 1c has a uniform light intensity distribution. As described above, part of the light emitted from the active layer 202 is trapped inside the light emitting device 1c by total internal reflection. The scattering pattern 293 scatters the incident light and changes a traveling direction of the scattered light. As described above, the scattering pattern 293 scatters the light captured inside the light emitting device 1c by total internal reflection, thereby changing the traveling direction of the light and allowing the light to be discharged from the light emitting device 1c. As a result, a light emitting device 1c may have improved light extraction efficiency.

[0077] Referring again to FIG. 5, the light emitting device 1c may further include an electrode pad (first electrode pad) 281. The first electrode pad 281 may include a conductive material. For example, the conductive material may include Al, Ti, Pt, Ag, Au, Pd, TiW, or various combinations thereof. The first electrode pad 281 functions as an electrode of the plurality of light emitting portions 210, that is, as an n-electrode in the embodiment. The first electrode pad 281 may be connected to a power line in a driving circuit of a display apparatus. FIG. 5 shows one first electrode pad 281, but the light emitting device 1c may include two or more first electrode pads 281.

[0078] FIG. 6 is a schematic cross-sectional view of a light emitting device 1d according to an embodiment. The light emitting device 1d of the embodiment is a flip-chip type light emitting device. FIG. 6 shows that the light emitting device 1d is transferred and flipped to the carrier substrate 110. The light emitting device 1d is attached to the carrier substrate 110 with the adhesive 112. The light emitting device 1d of the embodiment is different from the light emitting device 1c shown in FIG. 5 in that the light emitting device 1d includes a transmission conductive layer 294. Hereinafter, components that are the same as those of the light emitting device 1c shown in FIG. 5 are denoted by the same reference numerals, and differences from the light emitting device 1c are mainly described. Descriptions of the light emitting device 1, the light emitting device 1a, the light emitting device 1b, and the light emitting device 1c apply to the light emitting device 1d, unless contradictory.

[0079] Referring to FIG. 6, the light emitting device 1d further includes the transmission conductive layer 294. The transmission conductive layer 294 may be provided the surface 201a of the first conductive semiconductor layer 201 opposite to the active layer 202. The transmission conductive layer 294 may cover the surface 201a of the first conductive semiconductor layer 201 opposite to the active layer 202. The transmission conductive layer 294 may be formed in an area of the surface 201a of the first conductive

semiconductor layer **201** at least corresponding to the light emitting portion **210**. In the embodiment, the scattering pattern **293** is provided on the surface **201a** of the first conductive semiconductor layer **201**, and the transmission conductive layer **294** may be provided on the scattering pattern **293**. For example, the transmission conductive layer **294** may cover the scattering pattern **293**. The transmission conductive layer **294** may include a transparent conductive material. The transmission conductive layer **294** may function, for example, as an n-electrode. The first electrode pad **281** may be further formed on the transmission conductive layer **294**.

[0080] The transmission conductive layer **294** may function as a protective layer that protects the surface **201a** or the scattering pattern **293** of the first conductive semiconductor layer **201** when forming the via hole **291** and/or the via hole **292**. The transmission conductive layer **294** may include the same material as the second contact layer **220b**, for example, ITO. In this case, the transmission conductive layer **294** may be referred to as a protective layer.

[0081] The transmission conductive layer **294** may function as a refractive index matching layer that increases light transmittance by compensating for the difference in refractive index between the light emitting device **1d** and an external medium. The transmission conductive layer **294** may include a material having a refractive index between a refractive index of the epi structure **200** and a refractive index of an external medium, for example, air. For example, the transmission conductive layer **294** may include ITO.

[0082] The order in which the transmission conductive layer **294** and the first electrode pad **281** are stacked is not limited. FIG. 7 is a schematic cross-sectional view of a light emitting device **1e** according to an embodiment. As shown in FIG. 7, the first electrode pad **281** may be first formed on the surface **201a** of the first conductive semiconductor layer **201**, and the transmission conductive layer **294** may be provided on the surface **201a** of the first conductive semiconductor layer **201** and the first electrode pad **281**. For example, the transmission conductive layer **294** may cover the surface **201a** of the first conductive semiconductor layer **201** and the first electrode pad **281**.

[0083] FIGS. 8A to 8I are diagrams showing a method of manufacturing a light emitting device according to an embodiment. Embodiments of the method of manufacturing the light emitting device will be described with reference to FIGS. 8A to 8I.

[0084] Referring to FIG. 8A, the epi structure **200** including the first conductive semiconductor layer **201**, the active layer **202**, and the second conductive semiconductor layer **203** is formed on the growth substrate **100**.

[0085] The growth substrate **100** is, for example, a growth substrate for semiconductor single crystal growth, and may use a silicon (Si) substrate, a silicon carbide (SiC) substrate, a sapphire substrate, etc. Besides the above, a substrate including a material, suitable for the growth of the epi structure **200** to be formed on the substrate **100**, for example, AlN, AlGaIn, ZnO, GaAs, MgAl₂O₄, MgO, LiAlO₂, LiGaO₂, GaN, may be used. According to an embodiment, a buffer layer for epitaxial growth of the epi structure **200** may be provided on a surface of the substrate **100**, and the epi structure **200** may be grown on the buffer layer. A silicon substrate is used as the substrate **100** in the embodiment.

[0086] The epi structure **200** may be formed by sequentially growing the first conductive semiconductor layer **201**,

the active layer **202**, and the second conductive semiconductor layer **203** on the growth substrate **100**. In an embodiment, a buffer layer may be grown on the growth substrate **100**. For example, the buffer layer may be provided on the second conductive semiconductor layer **203**. The epi structure **200** may include a Group III-V nitride semiconductor material. The Group III-V nitride semiconductor material may include, for example, GaN, InGaIn, AlInGaIn, AlGaInP, etc. In the embodiment, the epi structure **200** includes a GaN-based semiconductor material. The first conductive semiconductor layer **201** may be, for example, an n-GaN layer doped with n-type impurities. Si, Ge, Se, Te, etc. may be used as n-type impurities. The active layer **202** is a layer that emits light by electron-hole recombination, and may have a single quantum well or a multi quantum well structure as described above. For example, a quantum well layer and a barrier layer may be paired in the form of InGaIn/GaN, InGaIn/InGaIn, InGaIn/AlGaIn, or InGaIn/InAlGaIn to form the quantum well structure, and a band gap energy may be controlled according to a composition ratio of indium (In) in a material layer including indium (In), and thus, an emission wavelength band may be adjusted. In the embodiment, the second conductive semiconductor layer **203** may be a p-GaN layer doped with p-type impurities. The epi structure **200** may be formed using HVPE, MBE, MOVPE, MOCVD, other known methods, or a combination thereof.

[0087] An operation of separating the light emitting portion **210** and the electrode portion **260** from each other is performed. The light emitting portion **210** and the electrode portion **260** may be separated from each other by forming the trench **250** in the epi structure **200** to expose the first conductive semiconductor layer **201**. Referring to FIG. 8A, a mask pattern **301** is formed on an upper surface of the epi structure **200**. An opening **301a** corresponding to the light emitting portion **210** and an opening **301b** corresponding to the electrode portion **260** are provided in the mask pattern **301**. The mask pattern **301** may be formed as, for example, a SiO₂ hard mask.

[0088] The trench **250** may be formed by etching the epi structure **200** through the openings **301a** and **301b**, as shown in FIG. 8B. After etching is completed, the mask pattern **301** is removed. By the trench **250**, the epi structure **200** may be divided into a plurality of mesa structures, for example, four mesa structures **310a**, **310b**, **310c**, and **310d**. The mesa structures **310a**, **310b**, and **310c** may have the same height. The mesa structures **310a**, **310b**, and **310c** may be the light emitting portion **210**, and the mesa structure **310d** may be the electrode portion **260**.

[0089] According to an embodiment, etching may be performed using a dry etching process or a wet etching process. The dry etching process may use, for example, inductively coupled plasma (ICP). The wet etching process may be performed, for example, using a potassium hydroxide (KOH) solution or a tetramethyl ammonium hydroxide (TMAH) solution as an etchant. However, the disclosure is not limited thereto, and as such, the trench **250** may be formed in another manner.

[0090] According to an embodiment, sidewalls of the mesa structures **310a**, **310b**, **310c**, and **310d** may not be perpendicular to a surface of the substrate **100**. As shown by a dashed line **209** in FIG. 8B, the sidewalls of the mesa structures **310a**, **310b**, **310c**, and **310d** may have a tapered shape with respect to the substrate **100**. During a process of etching the epi structure **200**, due to the difference in an etch

rate between the surface **201b** of the first conductive semiconductor layer **201** close to the mask pattern **301** and the surface **201a** of the first conductive semiconductor layer **201** far from the mask pattern **301**, the widths of the mesa structures **310a**, **310b**, **310c**, and **310d** may gradually increase from the surface **201b** to the surface **201a**. In other words, a width of the trench **250** may gradually decrease from the surface **201b** toward the surface **201a**. Inclinations of the sidewalls of the mesa structures **310a**, **310b**, **310c**, and **310d** may vary depending on the type of etching process and etching conditions. According to the wet etching process, the mesa structures **310a**, **310b**, **310c**, and **310d** with greater inclination angles of sidewalls may be obtained compared to the dry etching process.

[0091] As shown in FIG. 8C, the plurality of pores **270** are formed in the first conductive semiconductor layer **201**. The plurality of pores **270** may be formed by etching the first conductive semiconductor layer **201** of the mesa structures **310a**, **310b**, **310c**, and **310d** in a transverse direction inward through the trench **250**. For example, the plurality of pores **270** may be formed by a wet type electroetching process. Accordingly, in FIG. 8C, the plurality of pores **270** are shown in the form of holes and extending in the transverse direction from the sidewalls. The plurality of pores **270** may be formed at positions spaced apart from the active layer **202** by at least 100 nm. According to an embodiment, before forming the plurality of pores **270**, side surfaces of the mesa structures **310a**, **310b**, **310c**, and **310d**, excluding an area where the plurality of pores **270** are to be formed, may be masked by a protective mask. The protective mask may be formed to an area of at least 100 nm from a lower surface of the active layer **202**.

[0092] A process of forming the first contact layer **220a** and the second contact layer **220b** on the light emitting portion **210** and the electrode portion **260**, respectively, is performed. After forming a deposition mask exposing the upper surfaces of the mesa structures **310a**, **310b**, **310c**, and **310d** in the state shown in FIG. 8C, a material capable of forming an ohmic contact with the second conductive semiconductor layer **203** is deposited on the exposed upper surfaces of the mesa structures **310a**, **310b**, **310c**, and **310d**. Then, as shown in FIG. 8D, the first contact layer **220a** and the second contact layer **220b** are formed on the light emitting portion **210** and the electrode portion **260**, respectively. The first contact layer **220a** and the second contact layer **220b** may each include a conductive material. The conductive material may include Al, Ti, Pt, Ag, Au, Pd, TiW, or various combinations thereof. The first contact layer **220a** and the second contact layer **220b** may each include a transparent conductive material. The transparent conductive material may include ITO. The process of forming the first contact layer **220a** and the second contact layer **220b** is not limited. The first contact layer **220a** and the second contact layer **220b** may be formed, for example, using sputtering, atomic layer deposition (ALD), plasma enhanced atomic layer deposition (PEALD), chemical vapor deposition (CVD), plasma enhanced chemical vapor deposition (PECVD), physical vapor deposition (PVD), other known methods, or a combination thereof.

[0093] As is described below, the second contact layer **220b** may include a material that may function as an etch stop layer with respect to a semiconductor material forming the epi structure **200**, for example, GaN. For example, the second contact layer **220b** may include ITO. The first contact

layer **220a** and the second contact layer **220b** may include the same material, for example, ITO. As a result, the process may be simplified.

[0094] As shown in FIG. 8E, the passivation layer **230** may be provided on the side surfaces of the light emitting portion **210** and the electrode portion **260**, and the first contact layer **220a** and the second contact layer **220b**. The passivation layer **230** may be provided on the side surfaces of the mesa structures **310a**, **310b**, **310c**, and **310d** exposed by the trench **250**, the upper surface of the first conductive semiconductor layer **201** exposed by the trench **250**, and the side surfaces and upper surfaces of the first contact layer **220a** and the second contact layer **220b**. For example, the passivation layer **230** covering the side surfaces of the light emitting portion **210** and the electrode portion **260**, and the first contact layer **220a** and the second contact layer **220b** may be formed. The passivation layer **230** may cover the side surfaces of the mesa structures **310a**, **310b**, **310c**, and **310d** exposed by the trench **250**, the upper surface of the first conductive semiconductor layer **201** exposed by the trench **250**, and the side surfaces and upper surfaces of the first contact layer **220a** and the second contact layer **220b**. The passivation layer **230** may include a dielectric material. The dielectric material may include SiO₂, TiO₂, Si₃N₄, AlO_x, AlO_xN_y, Ta₂O₅, TiN, AlN, ZrO₂, TiAlN, TiSiN, HfO_x, or various combinations thereof. A thickness of the passivation layer **230** may be about 5 nm or more and about 50 nm or less, for example, about 10 nm or more and about 30 nm or less. The passivation layer **230** may have a single-layer structure or a multi-layer structure. The passivation layer **230** may include a material that may improve luminous efficiency according to an emission wavelength of the light emitting portion **210**. In an example case in which light emitting portion **210** emits red light, the passivation layer **230** may include AlO_xN_y, and in an example case in which the light emitting portion **210** emits blue light, the passivation layer **230** may include Ta₂O₅. A process of forming the passivation layer **230** is not limited. For example, the passivation layer **230** may be formed using sputtering, ALD, PEALD, CVD, PECVD, PVD, other known methods, or a combination thereof.

[0095] Referring to FIG. 8F, the opening **231** exposing the first contact layer **220a** is formed in the passivation layer **230**. For example, an etch mask may be formed on the passivation layer **230**. For example, the etch mask may cover the passivation layer **230**. Etch holes are provided in areas corresponding to the mesa structures **310a**, **310b**, and **310c** of the etch mask. The opening **231** is formed by etching the passivation layer **230** through the etch holes. The first contact layer **220a** is partially exposed through the opening **231**. The etch mask is removed.

[0096] As shown in FIG. 8G, the conductive reflective layer **240** may be provided on the passivation layer **230**. For example, the conductive reflective layer **240** covering the passivation layer **230** is formed. The conductive reflective layer **240** may include a conductive material. The conductive reflective layer **240** may include a conductive material having reflectivity. For example, the conductive material may include Al, Ti, Pt, Ag, Au, Pd, TiW, or various combinations thereof. The conductive reflective layer **240** may have a single-layer structure or a multi-layer structure. A process of forming the conductive reflective layer **240** is not limited. For example, the conductive reflective layer **240**

may be formed using sputtering, ALD, PEALD, CVD, PECVD, PVD, other known methods, or a combination thereof.

[0097] The conductive reflective layer **240** is in contact with the first contact layer **220a** of the light emitting portion **210** through the opening **231**. Because the passivation layer **230** is disposed between the conductive reflective layer **240** and the second contact layer **220b** of the electrode portion **260**, the conductive reflective layer **240** does not contact the second contact layer **220b** of the electrode portion **260**.

[0098] As shown in FIG. 8G, a process of forming the electrode pad (second electrode pad) **282** on the conductive reflective layer **240** may be further performed. The second electrode pad **282** may include a conductive material. For example, the conductive material may include Al, Ti, Pt, Ag, Au, Pd, TiW, or various combinations thereof. The second electrode pad **282** may be formed, for example, using sputtering, ALD, PEALD, CVD, PECVD, PVD, other known methods, or a combination thereof.

[0099] The light emitting device **1** shown in FIG. 1 may be manufactured through the above-described processes. Next, an embodiment of a method of manufacturing the light emitting device **1a** shown in FIG. 3 is described.

[0100] After the process shown in FIGS. 8A to 8G is performed, a process result, that is, the epi structure **200**, may be transferred to the carrier substrate **110**. The carrier substrate **110** may be, for example, a silicon substrate. The epi structure **200** may be attached to the carrier substrate **110** with the adhesive **112**. Then, the epi structure **200** is flipped. Then, as shown in FIG. 8H, in an example case in which the growth substrate **100** is separated, the surface **201a** of the first conductive semiconductor layer **201** of the epi structure **200** is exposed. According to an embodiment, a process of etching the exposed surface of the first conductive semiconductor layer **201** of the epi structure **200** may be added. For example, the exposed surface of the first conductive semiconductor layer **201** of the epi structure **200** may be etched to, for example, about 2 μm .

[0101] Next, an etch mask **303** including an etch hole **303a** is formed on the surface **201a** of the first conductive semiconductor layer **201**. The etch hole **303a** is formed in an area corresponding to the electrode portion **260**. The size of the etch hole **303a** may be less than the size of the electrode portion **260**. The etch mask **303** may include, for example, photoresist. The epi structure **200** of the electrode portion **260** is etched through the etch hole **303a**. Etching may be performed using a dry etching process or a wet etching process. The dry etching process may use, for example, ICP. The wet etching process may be performed, for example, using a KOH solution or a TMAH solution as an etchant. In the embodiment, the etching process is performed by the wet etching process using the TMAH solution as an etchant. In the etching process, the second contact layer **220b** including ITO functions as an etch stop layer. As a result, the via hole **291** penetrating the first conductive semiconductor layer **201**, the active layer **202**, and the second conductive semiconductor layer **203** of the electrode portion **260** may be formed, and the electrode portion **260** may be disabled (non-emitting). The etch mask **303** is removed.

[0102] In an example case in which an etching process error is $\pm 5\%$, and a GaN layer with a thickness of about 8 μm is etched, the maximum etch error of the GaN layer is about 800 nm. Such a large etch error may damage the conductive reflective film **240** which is used as a p-electrode.

According to experiments, an etch rate of GaN is about 8.2 nm/s, which is about 19.5 times an etch rate of ITO, which is about 0.42 nm/s. Mathematically, an ITO layer with a thickness of about 100 nm may correspond to a GaN layer with a thickness of about 2 μm in terms of etch thickness, and the second contact layer **220b** including ITO may function as an excellent etch stop layer. By forming the second contact layer **220b** with ITO of an appropriate thickness in consideration of the etching process error, damage to the conductive reflective layer **240** that is the p-electrode may be prevented in the etching process for forming the via hole **291** in the electrode portion **260**. As described above, the light emitting device **1a** shown in FIG. 3 including the via hole **291** may be manufactured.

[0103] Next, an embodiment of a method of manufacturing the light emitting device **1b** shown in FIG. 4 including the via hole **292** is described.

[0104] By the process shown in FIG. 8H, the first via hole **292a** penetrating the first conductive semiconductor layer **201**, the active layer **202**, and the second conductive semiconductor layer **203** of the electrode portion **260** may be formed. Then, as shown in FIG. 8I, an etch mask **304** including an etch hole **304a** is formed on the second contact layer **220b**. The etch mask **304** may include photoresist, for example. The second via hole **292b** is formed by etching the second contact layer **220b** and the passivation layer **230** through the etch hole **304a**. Etching may be performed using a dry etching process or a wet etching process. As a result, the via hole **292** including the first via hole **292a** penetrating the epi structure **200** of the electrode portion **260** and the second via hole **292b** penetrating the second contact layer **220b** and the passivation layer **230** to exposing the conductive reflective layer **240** may be formed. The etch mask **304** is removed.

[0105] As shown in FIGS. 5, 6, and 7, the scattering pattern **293** may be provided on the surface **201a** of the first conductive semiconductor layer **201**. A first electrode pad **281** may be provided on the surface **201a** of the first conductive semiconductor layer **201**. The transmission conductive layer **294** may be provided on a surface of the first conductive semiconductor layer **201**.

[0106] FIGS. 9A and 9B are diagrams showing an operation of forming the scattering pattern **293**, as an embodiment of a light emitting device manufacturing method. First, processes of FIGS. 8A to 8G are performed. Then, as shown in FIG. 9A, a process result, that is, the epi structure **200**, may be transferred to the carrier substrate **110**. The carrier substrate **110** may be, for example, a silicon substrate. The epi structure **200** may be attached to the carrier substrate **110** with the adhesive **112**. Then, the epi structure **200** is flipped. Then, when the growth substrate **100** is separated, the surface **201a** of the first conductive semiconductor layer **201** of the epi structure **200** is exposed. According to an embodiment, a process of etching the exposed surface of the first conductive semiconductor layer **201** of the epi structure **200** may be added. For example, the exposed surface of the first conductive semiconductor layer **201** of the epi structure **200** may be etched to, for example, about 2 μm .

[0107] Next, the scattering pattern **293** may be formed on the surface **201a** of the first conductive semiconductor layer **201**, as shown in FIG. 9A, by etching the surface **201a** of the first conductive semiconductor layer **201**. Etching may be performed, for example, using a wet etching process using a TMAH solution as an etchant. Next, the via hole **291** or the

via hole 292 may be formed as shown in FIG. 9B by performing the processes of FIGS. 8H and 8I.

[0108] According to an embodiment, the scattering pattern 293 may be formed on the surface 201a of the first conductive semiconductor layer 201 by performing a process of FIG. 9A after performing the process of FIGS. 8A to 8I. In this case, before an etching process for forming the scattering pattern 293 is performed, a protective layer blocking the via hole 292 to prevent the conductive reflective layer 240 from being damaged through the via hole 292 may be provided. After forming the scattering pattern 293, the protective layer is removed.

[0109] Next, in the state shown in FIG. 9B, a process of forming the first electrode pad 281 on the surface 201a of the first conductive semiconductor layer 201 may be performed, as shown in FIG. 5. The first electrode pad 281 may include a conductive material. For example, the conductive material may include Al, Ti, Pt, Ag, Au, Pd, TiW, or various combinations thereof. The first electrode pad 281 may be formed, for example, using sputtering, ALD, PEALD, CVD, PECVD, PVD, other known methods, or a combination thereof. A plurality of first electrode pads 281 may be formed. As a result, the light emitting device 1c shown in FIG. 5 may be manufactured.

[0110] Before forming the first electrode pad 281, that is, in the state shown in FIG. 9B, a process of forming the transmission conductive layer 294 on the surface 201a of the first conductive semiconductor layer 201 may be performed, as shown in FIG. 6. The transmission conductive layer 294 may include a transparent conductive material. For example, the transparent conductive material may include ITO. The transmission conductive layer 294 may be formed in an area of the surface 201a of the first conductive semiconductor layer 201 at least corresponding to the light emitting portion 210. According to an embodiment, the transmission conductive layer 294 may be provided on the scattering pattern 293. For example, the transmission conductive layer 294 may cover the scattering pattern 293. The transmission conductive layer 294 may be formed, for example, using ALD. Then, the first electrode pad 281 may be formed on the transmission conductive layer 294. As a result, the light emitting device 1d shown in FIG. 6 may be manufactured.

[0111] In the state shown in FIG. 9B, the first electrode pad 281 may be first formed on the surface 201a of the first conductive semiconductor layer 201 as shown in FIG. 5, and then, as shown in FIG. 7, the transmission conductive layer 294 may be provided on the surface 201a of the first conductive semiconductor layer 201 and the first electrode pad 281. For example, the transmission conductive layer 294 covering the surface 201a of the first conductive semiconductor layer 201 and the first electrode pad 281 may be formed. By this, the light emitting device 1e shown in FIG. 7 may be manufactured.

[0112] FIG. 10 is a schematic diagram of a display apparatus according to an embodiment. Referring to FIG. 10, the display apparatus may include a display panel 7110 and a controller 7160. The display panel 7110 may include a light emitting structure 7112 and a driving circuit 7115. The driving circuit 7115 may control an operation of the light emitting structure 7112. For example, the driving circuit 7115 may switch the light emitting structure 7112 on or off. The light emitting structure 7112 may include a plurality of light emitting devices described with reference to FIGS. 1 to 7 above. The plurality of light emitting devices may be

arranged, for example, in a two-dimensional array. The driving circuit 7115 may include a plurality of switching devices individually switching on-off the plurality of light emitting devices. The controller 7160 may input on-off switching signals of the plurality of light emitting devices to the driving circuit 7115 according to an image signal.

[0113] FIG. 11 is a block diagram of an electronic device 8201 including a display according to an embodiment. Referring to FIG. 11, the electronic device 8201 may be provided in a network environment 8200. In the network environment 8200, the electronic device 8201 may communicate with another electronic device 8202 through a first network 8298 (such as a short-range wireless communication network, and the like), or communicate with another electronic device 8204 and/or a server 8208 through a second network 8299 (such as a remote wireless communication network). The electronic device 8201 may communicate with the electronic device 8204 through the server 8208. The electronic device 8201 may include a processor 8220, a memory 8230, an input device 8250, an audio output device 8255, a display apparatus 8260, an audio module 8270, a sensor module 8276, and an interface 8277, a haptic module 8279, a camera module 8280, a power management module 8288, a battery 8289, a communication module 8290, a subscriber identification module 8296, and/or an antenna module 8297. In the electronic device 8201, some of these components may be omitted or other components may be added. Some of these components may be implemented as one integrated circuit. For example, the sensor module 8276 (fingerprint sensor, iris sensor, illuminance sensor, etc.) may be implemented by being embedded in the display apparatus 8260 (display, etc.).

[0114] The processor 8220 may execute software (the program 8240, etc.) to control one or a plurality of other components (such as hardware, software components, etc.) of the electronic device 8201 connected to the processor 8220, and perform various data processing or operations. As part of data processing or operation, the processor 8220 may load commands and/or data received from other components (the sensor module 8276, the communication module 8290, etc.) into the volatile memory 8232, process commands and/or data stored in the volatile memory 8232, and store result data in the nonvolatile memory 8234. The processor 8220 may include a main processor 8221 (such as a central processing unit, an application processor, etc.) and a secondary processor 8223 (such as a graphics processing unit, an image signal processor, a sensor hub processor, a communication processor, etc.) that may be operated independently or together. The secondary processor 8223 may use less power than the main processor 8221 and may perform specialized functions.

[0115] The secondary processor 8223 may control functions and/or states related to some of the components of the electronic device 8202 (such as the display apparatus 8260, the sensor module 8276, the communication module 8290, etc.) instead of the main processor 8221 while the main processor 8221 is in an inactive state (sleep state), or with the main processor 8221 while the main processor 8221 is in an active state (application execution state). The secondary processor 8223 (such as an image signal processor, a communication processor, etc.) may be implemented as part of other functionally related components (such as the camera module 8280, the communication module 8290, etc.).

[0116] The memory **8230** may store various data required by components of the electronic device **8201** (such as the processor **8220**, the sensor module **8276**, etc.) The data may include, for example, software (such as the program **8240**, etc.) and input data and/or output data for commands related thereto. The memory **8230** may include a volatile memory **8232** and/or a nonvolatile memory **8234**.

[0117] The program **8240** may be stored as software in the memory **8230** and may include an operating system **8242**, a middleware **8244**, and/or an application **8246**.

[0118] The input device **8250** may receive commands and/or data to be used for components (such as the processor **8220**, etc.) of the electronic device **8201** from outside (a user) of the electronic device **8201**. The input device **8250** may include a remote controller, a microphone, a mouse, a keyboard, and/or a digital pen (such as a stylus pen).

[0119] The audio output device **8255** may output an audio signal to the outside of the electronic device **8201**. The audio output device **8255** may include a speaker and/or a receiver. The speaker may be used for general purposes such as multimedia playback or recording playback, and the receiver may be used to receive incoming calls. The receiver may be combined as a part of the speaker or may be implemented as an independent separate device.

[0120] The display apparatus **8260** may visually provide information to the outside of the electronic device **8201**. The display apparatus **8260** may include the display, a hologram device, or a projector and a control circuit for controlling the device. The display apparatus **8260** may include the display described with reference to FIG. 10. The display apparatus **8260** may include a touch circuit set to sense a touch, and/or a sensor circuit (such as a pressure sensor) set to measure the strength of a force generated by the touch.

[0121] The audio module **8270** may convert sound into an electrical signal, or conversely, may convert an electrical signal into sound. The audio module **8270** may acquire sound through the input device **8250** or output sound through speakers and/or headphones of the audio output device **8255**, and/or another electronic device (such as the electronic device **8102**) directly or wirelessly connected to electronic device **8201**.

[0122] The sensor module **8276** may detect an operating state (such as power, temperature, and the like) of the electronic device **8201** or an external environmental state (such as a user state, and the like), and generate an electrical signal and/or data value corresponding to the detected state. The sensor module **8276** may include a gesture sensor, a gyro sensor, a barometric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, and/or an illuminance sensor.

[0123] The interface **8277** may support one or more specified protocols that may be used for the electronic device **8201** to connect directly or wirelessly with another electronic device (such as the electronic device **8102**). The interface **8277** may include a High Definition Multimedia Interface (HDMI), a Universal Serial Bus (USB) interface, an SD card interface, and/or an audio interface.

[0124] The connection terminal **8278** may include a connector through which the electronic device **8201** may be physically connected to another electronic device (such as the electronic device **8102**). The connection terminal **8278**

may include an HDMI connector, a USB connector, an SD card connector, and/or an audio connector (such as a headphone connector).

[0125] The haptic module **8279** may convert an electrical signal into a mechanical stimulus (such as vibration, movement, etc.) or an electrical stimulus that a user may perceive through a tactile or motor sense. The haptic module **8279** may include a motor, a piezoelectric element, and/or an electrical stimulation device.

[0126] The camera module **8280** may capture a still image and a video. The camera module **8280** may include a lens assembly including one or more lenses, image sensors, image signal processors, and/or flashes. The lens assembly included in the camera module **8280** may collect light emitted from a subject that is a target of image capturing.

[0127] The power management module **8288** may manage power supplied to the electronic device **8201**. The power management module **8388** may be implemented as a part of a Power Management Integrated Circuit (PMIC).

[0128] The battery **8289** may supply power to components of the electronic device **8201**. The battery **8289** may include a non-rechargeable primary cell, a rechargeable secondary cell, and/or a fuel cell.

[0129] The communication module **8290** may support establishing a direct (wired) communication channel and/or a wireless communication channel, and performing communication through the established communication channel between the electronic device **8201** and other electronic devices (such as the electronic device **8102**, the electronic device **8104**, the server **8108**, and the like). The communication module **8290** may include one or more communication processors that operate independently of the processor **8220** (such as an application processor) and support direct communication and/or wireless communication. The communication module **8290** may include a wireless communication module **8292** (such as a cellular communication module, a short-range wireless communication module, a Global Navigation Satellite System (GNSS) communication module, and the like) and/or a wired communication module **8294** (such as a local area network (LAN) communication module, a power line communication module, and the like). Among these communication modules, a corresponding communication module may communicate with other electronic devices through a first network **8298** (a short-range communication network such as Bluetooth, WiFi Direct, or Infrared Data Association (IrDA)) or a second network **8299** (a cellular network, the Internet, or a telecommunication network such as a computer network (such as LAN, WAN, and the like)). These various types of communication modules may be integrated into one component (such as a single chip, and the like), or may be implemented as a plurality of separate components (a plurality of chips). The wireless communication module **8292** may check and authenticate the electronic device **8201** in a communication network such as the first network **8298** and/or the second network **8299** using the subscriber information (such as international mobile subscriber identifier (IMSI), etc.) stored in the subscriber identification module **8296**.

[0130] The antenna module **8297** may transmit signals and/or power to the outside (such as other electronic devices) or receive signals and/or power from the outside. The antenna may include a radiator made of a conductive pattern formed on a substrate (such as PCB, etc.) The antenna module **8297** may include one or a plurality of

antennas. In an example case in which multiple antennas are included, an antenna suitable for a communication method used in a communication network such as the first network **8298** and/or the second network **8299** may be selected from the plurality of antennas by the communication module **8290**. Signals and/or power may be transmitted or received between the communication module **8290** and another electronic device through the selected antenna. In addition to the antenna, other components (such as RFIC) may be included as part of the antenna module **8297**.

[0131] Some of the components are connected to each other and may exchange signals (such as commands, data, and the like) through communication method between peripheral devices (such as bus, General Purpose Input and Output (GPIO), Serial Peripheral Interface (SPI), Mobile Industry Processor Interface (MIPI), and the like).

[0132] The command or data may be transmitted or received between the electronic device **8201** and the external electronic device **8204** through the server **8108** connected to the second network **8299**. The other electronic devices **8202** and **8204** may be the same or different types of devices as or from the electronic device **8201**. All or some of the operations executed by the electronic device **8201** may be executed by one or more of the other electronic devices **8202**, **8204**, and **8208**. In an example case in which the electronic device **8201** needs to perform a certain function or service, instead of executing the function or service itself, the electronic device **8201** may request one or more other electronic devices to perform the function or part or all of the service. One or more other electronic devices that receive the request may execute an additional function or service related to the request, and transmit a result of the execution to the electronic device **8201**. To this end, cloud computing, distributed computing, and/or client-server computing technology may be used.

[0133] The electronic device **8201** described above may be applied to various devices. According to functions of devices, various components of the electronic device **8201** described above may be appropriately modified, and components appropriate for performing the functions of the devices may be added. Application examples of the electronic device **8201** are described below.

[0134] FIG. 12 illustrates an embodiment of a mobile device **9100** as an application example of an electronic device. The mobile device **9100** may include a display apparatus **9110**. The display apparatus **9110** may include the display apparatus described with reference to FIG. 10. The display apparatus **9110** may have a foldable structure, for example, a multi-foldable structure.

[0135] FIG. 13 illustrates an embodiment of a vehicle head-up display apparatus **9200** as an application example of an electronic device. The vehicle head-up display apparatus **9200** may include a display **9210** provided in an area of a vehicle, and a light path changing member **9220** that converts an optical path so that a driver may see the image generated on the display **9210**. The display **9210** may include the display apparatus described with reference to FIG. 10.

[0136] FIG. 14 illustrates an embodiment of augmented reality glasses **9300** or virtual reality glasses **9300** as an application example of an electronic device. The augmented reality glasses (virtual reality glasses) **9300** may include a projection system **9310** that forms an image, and an element **9320** that guides the image from the projection system **9310**

into the user's eye. The projection system **9310** may include the display apparatus described with reference to FIG. 10.

[0137] FIG. 15 illustrates an embodiment of a signage **9400** as an application example of an electronic device. The signage **9400** may include the display apparatus described with reference to FIG. 10. The signage **9400** may be used for outdoor advertisement using a digital information display, and may control advertisement contents and the like through a communication network. The signage **9400** may be implemented, for example, through the electronic device described with reference to FIG. 11.

[0138] FIG. 16 illustrates an embodiment of a wearable display **9500** as an application example of an electronic device. The wearable display **9500** may include the display apparatus described with reference to FIG. 11. The wearable display **9500** may be implemented through the electronic device described with reference to FIG. 11.

[0139] The light emitting device according to the embodiment or the display apparatus including the light emitting device may also be applied to various products such as a rollable TV and a stretchable display.

[0140] According to some embodiments, the light emitting device with improved luminous efficiency and the display apparatus employing the light emitting device may be implemented.

[0141] According to some embodiments, the risk of damaging the electrode may be reduced in the process of non-emitting the electrode portion and exposing the electrode.

[0142] It should be understood that embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each embodiment should typically be considered as available for other similar features or aspects in other embodiments. While one or more embodiments have been described with reference to the figures, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope as defined by the following claims.

What is claimed is:

1. A light emitting device comprising:

a light emitting portion and an electrode portion, each of the light emitting portion and the electrode portion comprising:

an epi structure comprising:

a first conductive semiconductor layer comprising a plurality of pores,

an active layer, and

a second conductive semiconductor layer,

a trench between the light emitting portion, and the electrode portion to expose the first conductive semiconductor layer;

a first contact layer on the second conductive semiconductor layer of the light emitting portion;

a second contact layer on the second conductive semiconductor layer of the electrode portion;

a passivation layer provided on one or more side surfaces of the light emitting portion, and the electrode portion, and on the first contact layer and the second contact layer, the passivation layer comprising an opening exposing the first contact layer of the light emitting portion; and

a conductive reflective layer provided on the passivation layer and contacting the first contact layer of the light emitting portion through the opening.

2. The light emitting device of claim 1, wherein the plurality of pores are spaced apart from the active layer by at least 100 nm.

3. The light emitting device of claim 1, further comprising:

an electrode pad provided on the conductive reflective layer at a position corresponding to the electrode portion.

4. The light emitting device of claim 1, wherein the second contact layer comprises Indium Tin Oxide (ITO).

5. The light emitting device of claim 4, wherein a via hole penetrating the epi structure is formed on the electrode portion to expose the second contact layer.

6. The light emitting device of claim 5, wherein the via hole penetrates the second contact layer and the passivation layer to expose the conductive reflective layer.

7. The light emitting device of claim 1, further comprising:

an electrode pad provided on a surface of the first conductive semiconductor layer opposite to the active layer.

8. The light emitting device of claim 1, further comprising:

a transmission conductive layer provided on a surface of the first conductive semiconductor layer opposite to the active layer.

9. The light emitting device of claim 8, wherein the transmission conductive layer comprises Indium Tin Oxide (ITO).

10. The light emitting device of claim 1, further comprising:

a scattering pattern provided on a surface of the first conductive semiconductor layer opposite to the active layer.

11. A method of manufacturing a light emitting device, the method comprising:

forming an epi structure comprising a first conductive semiconductor layer, an active layer, and a second conductive semiconductor layer on a growth substrate; separating a light emitting portion and an electrode portion from each other by forming a trench in the epi structure to expose the first conductive semiconductor layer;

forming a first contact layer on the second conductive semiconductor layer of the light emitting portion, and forming a second contact layer on the second conductive semiconductor layer of the electrode portion;

forming a plurality of pores in the first conductive semiconductor layer;

forming a passivation layer on one or more side surfaces of the light emitting portion and the electrode portion, and on the first contact layer and the second contact layer;

forming an opening in the passivation layer to expose the first contact layer of the light emitting portion; and

forming a conductive reflective layer covering the passivation layer and contacting the first contact layer of the light emitting portion through the opening.

12. The method of claim 11, wherein the plurality of pores are spaced apart from the active layer by at least 100 nm.

13. The method of claim 11, further comprising:

forming an electrode pad on the conductive reflective layer at a position corresponding to the electrode portion.

14. The method of claim 11, wherein the first contact layer and the second contact layer each comprise Indium Tin Oxide (ITO).

15. The method of claim 11, further comprising:

transferring the epi structure onto a carrier substrate; and forming a via hole penetrating the first conductive semiconductor layer, the active layer, and the second conductive semiconductor layer of the electrode portion by using the second contact layer as an etch stop layer.

16. The method of claim 15, wherein the via hole penetrates the second contact layer and the passivation layer to expose the conductive reflective layer.

17. The method of claim 11, further comprising:

transferring the epi structure onto a carrier substrate; and forming an electrode pad on a surface of the first conductive semiconductor layer.

18. The method of claim 11, further comprising:

forming a scattering pattern on the surface of the first conductive semiconductor layer.

19. The method of claim 11, further comprising:

forming a transmission conductive layer on the surface of the first conductive semiconductor layer.

20. A display apparatus comprising:

a display panel comprising a plurality of light emitting devices, and a driving circuit configured to switch the plurality of light emitting devices on or off; and

a controller configured to input switching signals of the plurality of light emitting devices to the driving circuit according to an image signal,

wherein each of the plurality of light emitting devices comprises:

a light emitting portion and an electrode portion, each of the light emitting portion and the electrode portion comprising:

an epi structure comprising:

a first conductive semiconductor layer comprising a plurality of pores,

an active layer, and

a second conductive semiconductor layer,

a trench between the light emitting portion, and the electrode portion;

a first contact layer on the second conductive semiconductor layer of the light emitting portion;

a second contact layer on the second conductive semiconductor layer of the electrode portion;

a passivation layer provided on one or more side surfaces of the light emitting portion, and the electrode portion, and on the first contact layer and the second contact layer, the passivation layer comprising an opening exposing the first contact layer of the light emitting portion; and

a conductive reflective layer provided on the passivation layer and contacting the first contact layer of the light emitting portion through the opening.

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